

GLENCORE





Asset Management Framework

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A. INTRODUCTION

This Asset Management Framework is the foundational document for improving Asset Management Practices and subsequent delivery of unrivalled Asset Performance within Glencore Copper.

By definition, Asset Management is the coordinated activity of an organisation to realise value from its assets. (Source: ISO 55000, 2014 – International Organisation for Standardisation).

Maximising the value from our assets depends on the capability of our people as well as the business processes and systems that manage our assets.

This Framework exists to support the achievement of our Asset Management goals and objectives. The following principles define and direct our approach to managing our assets through:

- A dedication to deliver fit-for-purpose plant and equipment to our operations;
- A risk-based approach to all phases of the asset life cycle;
- Developing our people's competency for Asset Management;
- Creating effective work environments;
- Effective consultation, communication and decision making; and
- Incorporating continuous improvement into everything we do.

This document describes the Asset Management Framework and its underlying leading practices by which Glencore Copper manages asset performance, costs and risks towards achievement of its Asset Management purpose and vision. It contains:

- The Glencore Copper Asset Management Policy
- The purpose and vision of Asset Management;
- The scope of Asset Management;
- The delivery of Asset Management in Glencore Copper; and
- The intent, mind maps, drivers, outputs and key requirements of the Asset Management Framework Elements

B. ASSET MANAGEMENT PURPOSE AND VISION

The Glencore Copper definition of Asset Management Purpose and Vision underpins all of our Asset Management processes and activities, and applies to all levels and functions within our business.

It outlines the leadership, guidance and support processes provided by Glencore Copper to assist our sites to achieve their Asset Management requirements that contribute to the overall success of Glencore Copper.

B.1 Asset Management Purpose

To create value to shareholders by optimising reliability, eliminating losses, and maximising utilisation and asset life in a safe, cost effective manner.

B.2 Asset Management Vision

We have a culture that is motivated to present well-designed, safe, reliable equipment that is fit for purpose and consistently meets or exceeds nameplate capacity.

We pride ourselves on the elimination of failures while providing industry leading cost-effective reliability solutions to our operations.

We will not compromise on our standards and will be consistent with expectations for our people, suppliers and service providers.

C. SCOPE OF ASSET MANAGEMENT

C.1 Physical Asset Portfolio

The assets considered in the Asset Management Framework includes all physical assets associated with the extraction, beneficiation and transportation of the Copper commodity, which includes:

- Underground mobile and fixed plant and supporting infrastructure;
- Surface mobile and fixed plant and supporting infrastructure;
- Concentrators;
- Smelters;
- Refineries (SX/EW & Electrolytic);
- Bulk Storage and loading facilities;
- Utilities supporting mining, processing, and materials handling;
- Rail infrastructure and rolling stock (Glencore Copper owned);
- Buildings, workshops, warehouses, support equipment, tools; and
- Equipment spares and other items and services required to support these assets.

The scope includes all the data, resources, activities, processes, procedures, and systems required to support all phases of the life cycle of these assets from identification of need to asset disposal.

C.2 Scope of Activities

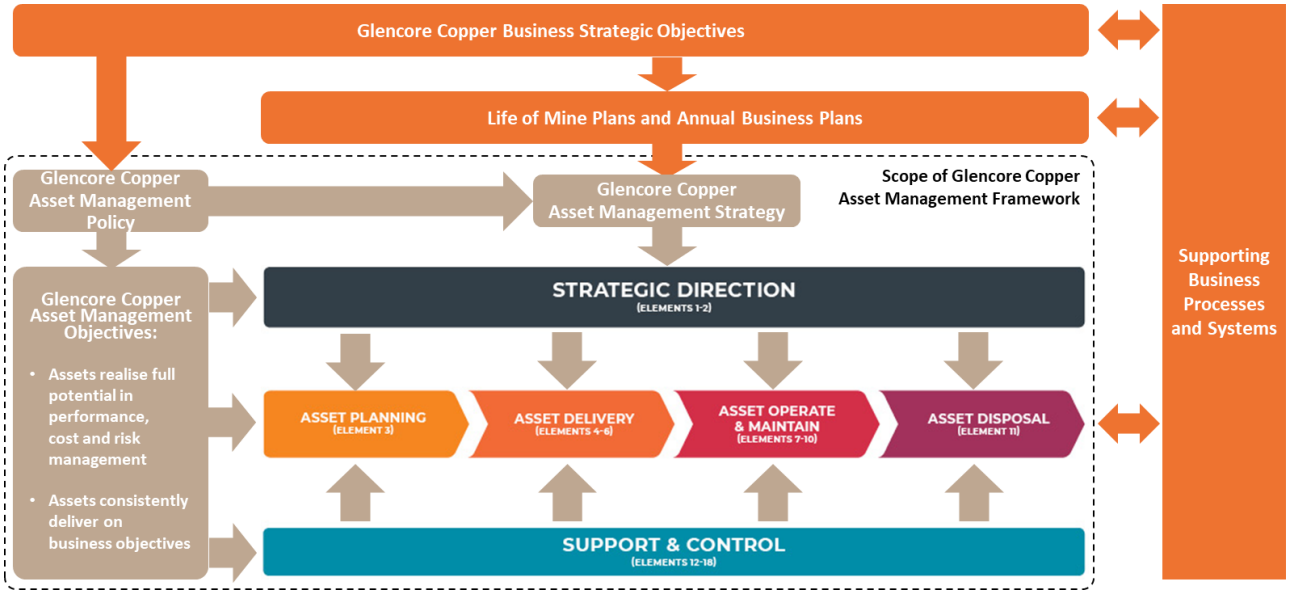
Glencore Copper manages physical assets in three specific ways all of which are covered by the Asset Management Framework. The alternative arrangements are listed below with their extent of coverage:

Operational Arrangement	Extent of Coverage by Asset Management Framework
As an owner, operator and maintainer of plant	All Asset Management activities covered
As an owner and operator in conjunction with a contracted maintenance service provider	All Asset Management activities covered
As an owner of plant in conjunction with contracted operations and maintenance service providers	All Asset Management activities covered

D. ASSET MANAGEMENT DELIVERY IN GLENCORE COPPER

Figure 1 describes the high-level Asset Management model, which enables Glencore Copper to deliver its Asset Management purpose and vision. Each component of the model is described below in further detail.

Figure 1: Asset Management Delivery in Glencore Copper



The scope of this Asset Management Framework is shown by the dotted line in **Figure 1** and encompasses:

- Glencore Copper Asset Management Policy;
- Glencore Copper Asset Management Strategy;
- Glencore Copper Asset Management Framework, comprising of eighteen elements;
- Each operation's Strategic Direction in relation to Asset Management;
- The Asset Planning, Asset Delivery, Asset Operate and Maintain, and Asset Disposal phases, implemented at each operation; and
- The Support and Control processes and systems implemented at each operation.

D.1 Development of Glencore Copper Business Strategic Objectives

Glencore Copper regions determine their strategic business direction based on the overall strategy and context set by Glencore Copper. The regions are required to develop plans to meet agreed targets based on individual operation capacities. The region's business strategic plans are updated annually and have a 4-year rolling outlook.

It is a requirement that each operation develop Life of Mine plans and accompanying Asset Management goals and Asset performance targets. In order to support this, an Asset Management Policy and Strategy has been developed and described in **Section D.3 and D.4**.

D.2 Development of Glencore Copper Asset Management Goals

Glencore Copper operations are required to produce Life of Mine (LOM) plans which are then rolled up to provide the Glencore Copper regions contribution to the overall Glencore Copper strategic objectives. LOM plans are used to determine the required future asset capacity and performance, which allow decisions to be made concerning:

- Addition of new assets to add capacity and/or replace unreliable, uneconomic, or obsolete assets;
- Refurbishment programmes for existing assets where condition is inadequate and it is economic to return them to design capability;
- Asset sustainment costs (labour and materials);
- Introduction of enhanced technology to improve effectiveness and efficiency;
- Development of Asset Management capabilities and competencies; and
- Minor and major projects to improve asset performance (availability and reliability).

The consideration of asset requirements and setting of goals in the Business Strategy and LOM plan establishes a 'line of sight' between Asset Management and the overall corporate objectives.

D.3 Glencore Copper Asset Management Policy

The management of physical assets over their life cycles is a complex task because of the need to consider the interfaces described in **Figure 1: Scope of Asset Management**. In the light of this complexity, Glencore Copper has established an Asset Management Policy, which provides high-level guidance for the business across all Asset Management activities.

D.4 Asset Management Strategy

Glencore Copper develops and refines the Asset Management Strategy in line with the Asset Management Policy and the overall business strategic objectives set by the Glencore Copper business. The process is aligned with the Glencore Copper Annual Business Planning budget submissions.

The Asset Management Strategy has a forward outlook of four years and is reviewed formally with each site on an annual basis. Sites are expected to roll out the strategy incorporating the Asset Management Framework Elements 1 to 18, described in **Section D.5** below.

D.5 Glencore Copper Asset Management Framework

The Asset Management Framework (AMF) consists of 18-elements. Each element is defined through an intent, mind map, drivers, outputs and key requirements.

Figure 2: Glencore Copper Asset Management Framework and the Eighteen Elements



D.6 Development of Site Strategic Direction

Each operation is responsible for developing its individual Asset Management objectives, budgets and plans in response to the LOM plans and in line with Glencore Copper's Asset Management Strategy. Elements 1 and 2 of the Asset Management Framework provide the key requirements.

Typical activities include:

- Development of site asset management governance structures including identification of the asset and asset system ownership, roles and responsibilities;
- Risk assessment of factors affecting the ability of the site to meet asset management goals and objectives and identification of appropriate controls;
- Contingency plans to cover for asset failure and incidents affecting operational output;
- Establishment of a productive work environment with asset management activities mapped to organisational roles;
- Creation of fully resourced initiatives to develop asset management capabilities and improve asset performance;
- Development of annual budgets which support the achievement of objectives; and
- Regular monitoring and reviews of progress against plan.

D.7 Asset Lifecycle Phases

Leading practice Asset Management requires clearly defined Asset Management processes to be in place at each operation. In order to define these processes consistently, Glencore Copper has defined the four-phase lifecycle model shown in **Figure 1 and Figure 2**. These processes enable the Site Asset Management Objectives, Budgets and Plans to be delivered and are supported by the Asset Management Support Processes.

Asset Planning: Asset Planning covers the assessment of existing asset capabilities, the development of asset strategies, objectives and plans, the establishment of the capabilities needed to deliver the plans and the review and forecasting process.

Asset Delivery: Utilising the processes in this phase, Glencore Copper develops its capability (systems, people and equipment). Documents equipment operating and maintenance strategies including energy, consumables, spares, reliability modelling and life cycle costs. Confirms all asset information is complete, and any new or modified assets are commissioned and handed over to operations effectively prior to the Operate and Maintain phase.

Asset Operation and Maintenance: Assets are operated within defined limits and are maintained against defined equipment strategies in accordance with effective work management processes. Longer duration maintenance is managed through a defined shutdown and overhaul process. Where maintenance activities are outsourced, these are managed in line with the Glencore Copper Asset Management Framework.

Asset Disposal: At the end of their useful lives, assets are dismantled, stored or disposed of in accordance with regulatory requirements and Glencore Copper policies and guidelines.

The asset lifecycle phases described above rely significantly on Support and Control processes and systems. These processes and systems are key enablers for the effective and efficient execution of the asset lifecycle phases.

E. FRAMEWORK STRUCTURE

The Asset Management Framework is made of eighteen (18) Elements. There is a relationship between all of the elements, as shown in **Figure 2**, however each element can be read and considered independent of the others.

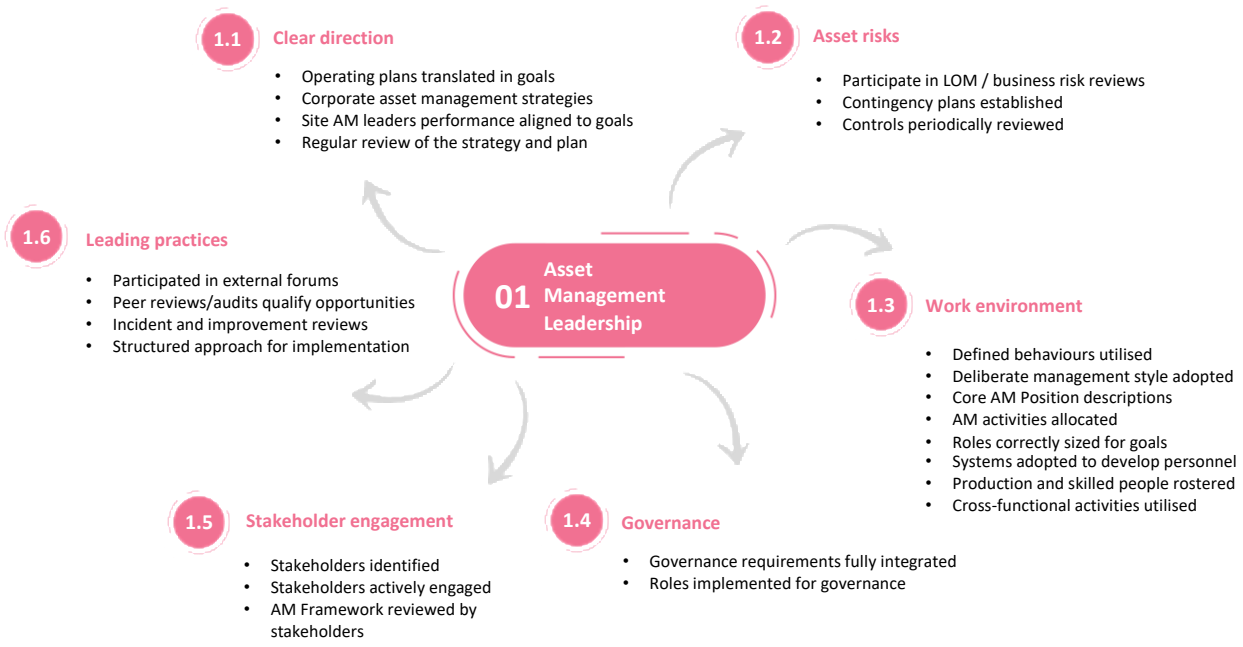
Each element is broken up into five (5) areas:

- **Intent:** The overall intent of how this element contributes to asset management;
- **Mind Map:** A visual depiction which shows the relationship between the Element (centre), Key requirement (1st point out from centre), criteria (2nd point out) and in some cases also includes guidance (3rd point out);
- **Drivers:** The key reasons why this element is included and the typical impacts it has on asset performance. Drivers are provided as information only to support the communication of the AMF;
- **Outputs:** Describes what is delivered as a result of achieving the requirements of this element; and
- **Key Requirements:** How the intent and outputs of the element are to be achieved. Each Key Requirement has a unique number referenced to the element. In nearly every case, sub-points indicate criteria, which support the achievement of that key requirement.

1. Asset Management Leadership

Intent

Site asset management leaders set the direction and are committed to resource and support Asset Management practices to effectively deliver the business strategic objectives throughout the operation's life.



Drivers:

- Appreciating that, for asset intensive copper businesses, plant and equipment performance are fundamental to business performance. Asset management leadership is critical.
- Site asset management leaders establish a clear long-term direction to suit present and future asset management needs.
- Existing clear governance guidelines and responsibilities limit asset management risks to tolerable levels.
- Applying good leadership means people are more effective in their roles within an environment of working effectively together and allowing capability development. People understand the link between their day-to-day asset management activities and strategic business objectives.
- Understanding that business systems effectiveness in managing physical assets impacts significantly on achieving asset management goals.
- Recognising the importance of leadership in both Glencore Sustainable Development standards and public standards such as ISO 55000.

1. Asset Management Leadership (continued)

Outputs

- Personnel are able to make decisions on the management of assets with an understanding of what is required from the assets in both the short and long term.
- Asset management related risks are kept to tolerable levels.
- Clearly defined roles, responsibilities and accountabilities for all aspects of asset management.
- Key Performance Indicators (KPI's) and targets are in place, results are analysed, shortfalls actioned and reviewed.
- Asset management governance is in place which reconciles conflicts of interest between stakeholders and is sustainable through changes in the organisation.
- External and internal stakeholder requirements are understood and met.
- Key learnings and best practices from other Glencore Copper operations are captured and implemented.

Key Requirements:

1.1 – Establishing and communicating a clear direction for asset management at each site demonstrates 'line of sight' between business objectives and asset management activities:

- **Translating** the site **operating plans to** site asset management **goals** to develop clear asset management strategy and plans.
- Translating **corporate asset management strategies** into site asset management strategies to achieve the required asset management capability.
- **Aligning site** asset management **leaders**' personal performance **and** site asset management **goals**.
- Establishing and **regularly reviewing** site asset management **strategy and plans** to confirm delivery of site asset management goals.

1.2 – Reviewing and managing asset risks to ensure all risks are tolerable:

- Asset management leaders **participate in** a site's annual **Life of Mine/Operation (LOM) and critical business risk review** with the requirements actioned within agreed timeframes.
- **Establishing** asset management **contingency plans** for significant risks arising from critical business risk reviews.
- Existing asset management **controls** for any significant risk identified in the critical business risk review are **reviewed periodically** for adequacy and functionality.

1.3 – Fostering work environments for people to effectively perform their responsibilities at all phases of an asset's lifecycle:

- **Utilising clearly** defined asset management **behaviours** in recruitment, performance appraisals and supervision.
- **Adopting a** consistent and **deliberate management style** and approach to changes.
- **Defining position descriptions** to include core asset management accountability and responsibility.
- **Allocating** asset management **activities** to ensure clear authority and responsibilities.
- **Supporting** asset management **goals with** manageable **role sizes** and responsibilities.
- **Adopting systems to develop** asset management **personnel**.
- **Rostering production** schedules **and skilled people** to support asset management goals.
- **Using cross-functional activities** to support asset management decision-making.

1. Asset Management Leadership (continued)

1.4 – Establishing governance processes for all aspects of the management of assets supported by documented standards, procedures and guidelines:

- **Integrating the requirements of** Glencore Copper, **regulatory legislation**, external standards and guidelines into site asset management processes and systems.
- **Implementing roles**, responsibilities and process where **governance** arrangements are required. (For projects, this includes sponsorship, steering groups and approval-gates).

1.5 – Planning and executing communications to ensure stakeholder engagement and shared commitment throughout the asset lifecycle and across all functions:

- **Identifying** and actively managing asset management **stakeholders**.
- **Working** with asset management **stakeholders** to achieve asset management objectives across an asset's lifecycle.
- **Reviewing** processes and outcomes for all elements of **the Asset Management Framework**.

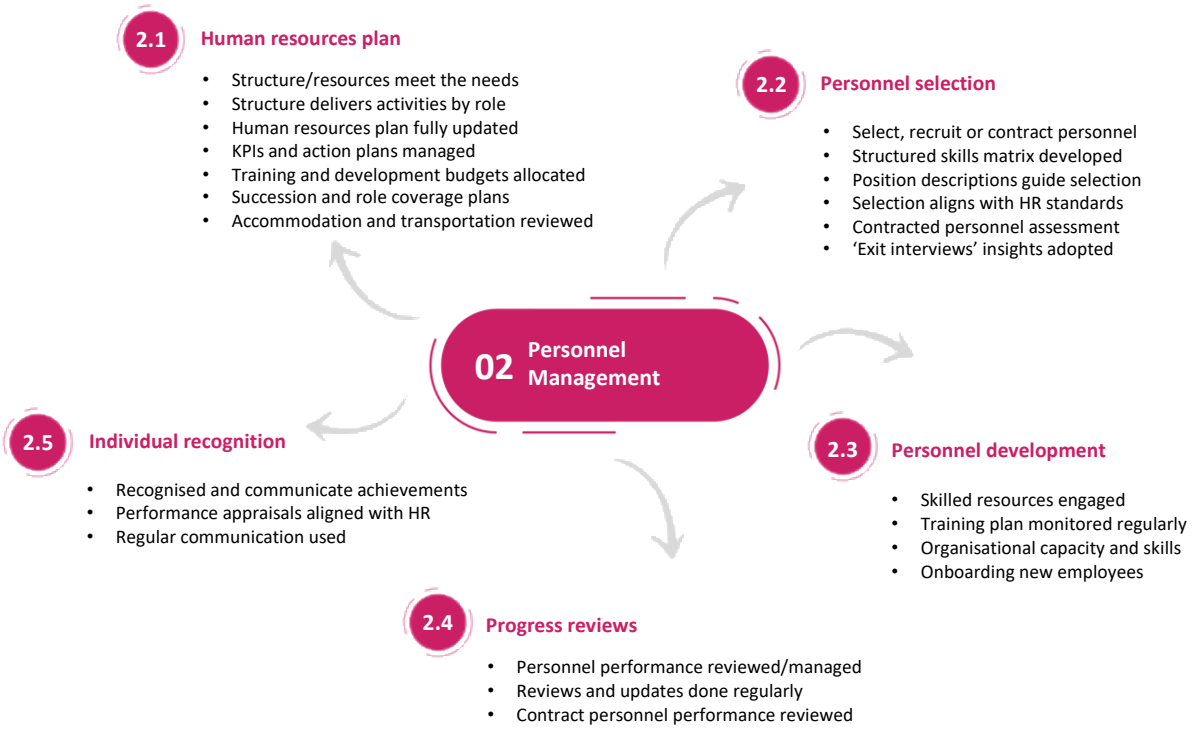
1.6 – Promoting a proactive approach, identifying leading asset management practices and implementing lessons learned from significant events:

- **Participating in external forums**, legislator meetings, conferences and industry groups.
- Qualifying opportunity areas using **peer reviews and audits**.
- Monitoring **incident and improvement reviews** by others with application at site.
- Applying a **structured approach to** recording participation, feedback and **actions implemented**.

2. Personnel Management

Intent

Attract, develop and retain sufficient personnel with the appropriate competencies, experience and behaviours to deliver the present and future asset management strategies.



Drivers:

- Applying the Asset Management Framework is critical to achieve the business goals and objectives, relying heavily on people bringing together all the asset management tasks, processes and standards (what, when and how).
- Recognising that human resources, including contractors, are a large percentage of total expenditure.
- Understanding people's competency, experience and behaviour impact significantly on commitments to Health, Safety, Environment and Community (HSEC) and operational performance.
- Ensuring asset owners provide Human Resources (HR) personnel with accurate and relevant inputs (i.e. behaviour, experience, competency) to deliver the right results.
- Understanding that asset management relies heavily on people (internal and external) at all levels in the management structure and so a key competency is an ability to manage people.

2. Personnel Management (continued)

- Enabling people development so operating, technical and maintenance tasks are completed safely and meet quality requirements.
- Using clear experience, competency and behavioural standards to delivery asset management performance.
- Integrated skills and competency matrix.

Outputs:

- Deploy knowledgeable and competent workforce at each site.
- Efficient and safe activities.
- Systems in place to ensure people achieve their desired potential and employee turnover is minimised.
- People are recognised for delivery of asset management performance.
- Contractors supply people that meet the site asset management standards.
- Accurate forecast and plan for the selection and development of people to meet asset management strategies.

Key Requirements:

2.1 – Structuring and resourcing the current and future asset management needs to consistently meet business objectives through an effective human resource plan:

- Forecasting **resourcing** levels.
- Identifying required changes to organisation **structure**.
- Changing rosters and shift patterns.
- Reviewing contracted activities.
- Changing operating and maintenance strategies.
- Reviewing management processes for new roles or changes to existing roles.
- Mapping asset management activities to roles.
- Using gap analysis against existing incumbents and a risk-based action plan.
- **Managing** individual **KPIs and action plans**.
- **Allocating training and development** budgets.
- Developing **succession and role coverage plans**.
- **Reviewing accommodation and transportation** arrangements as needed.

2.2 – Selecting, recruiting or contracting asset management personnel to deliver asset management objectives and to sustain an effective work environment:

- Aligning with Glencore Copper HR standards.
- Including **selection** criteria based on defined asset management competencies, experience and behaviours.
- Basing pre-defined **roles** with position descriptions **meeting Glencore HR standards** and incorporating site asset management arrangements.
- Awarding **contracts using an assessment** of asset management competency, experience and behaviours.
- **Adopting 'exit interviews'** to identify selection process and criteria improvement opportunities.

2. Personnel Management (continued)

2.3 – Developing people to meet current and future business needs and agreed individual development plans:

- Monitoring and updating training plans.
- Supervising and mentoring arrangements for developing personnel, such as:
    - Managing trainees, graduates, apprentices.
    - Changing between process teams or disciplines.
    - Promoting personnel.
    - Allocating additional duties or expanding roles.
    - Rotating people between business functions to achieve planned capability.
    - Ensuring up-to-date records of progress and achieved competence.
    - Ensuring role or responsibility changes under the resource plan are implemented and reasons clearly communicated.

2.4 – **Reviewing and managing personnel performance** and progress towards achieving asset management objectives:

- Developing **personnel** through informal asset management **performance reviews** and initiatives.
- **Reviewing and updating progress regularly** against the Resource Plan, such as:
    - Succession plans.
    - Competency requirements.
    - Planned resource levels.
    - Capability of contracted resources.
    - Role responsibilities and functional spreads.
- **Reporting contract personnel performance**.

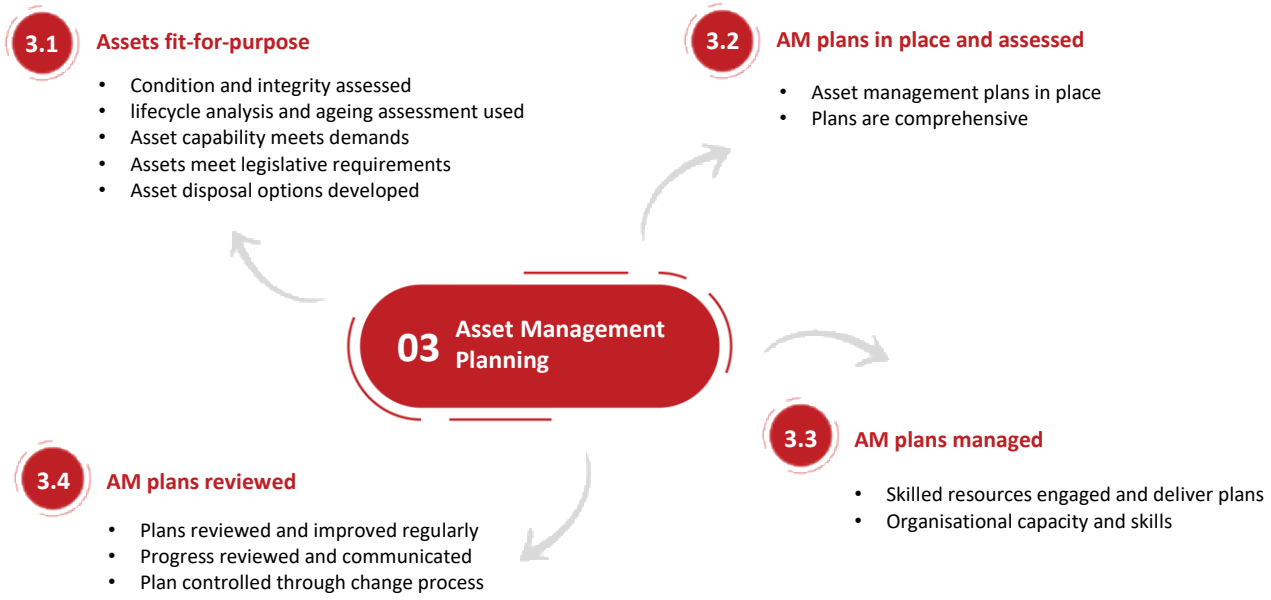
2.5 – **Recognising and communicating** individual development to support the **achievement** of asset management objectives.

- **Aligning performance appraisals with HR requirements** linking rewards and recognition to both business and asset management outcomes.
- Developing **regular communication channels** and forums to provide feedback on achievements.

3. Asset Management Planning

Intent

Asset management plans and initiatives are effectively developed to ensure the assets meet the short, medium and long-term business objectives.



Drivers:

- Reviewing business goals, objectives and critical risks at least annually, using the development of asset management activities to achieve goals and objectives.
- Recognising changes in asset demand may require new or modified business systems, processes and people competencies to deliver the required performance.
- Using critical asset lifecycle costing, operability and maintainability is critical for evaluating asset investment.
- Developing improved asset management solutions with defined and understood business requirements, due diligence and options analysis for meeting requirements.
- Deciding on good, timely asset replacement or upgrades by utilising lifecycle costings, performance trends, regulatory requirements and new technology opportunities.
- Planning processes recognise that asset and system implementations have long lead times before they start to deliver effective solutions.
- Understanding the importance of linking asset planning to the organisation's strategic objectives as specified by ISO 55000, and other asset management process models.

3. Asset Management Planning (continued)

Outputs:

- The submitted LOM/Life of Operations Business Plan contains all the asset requirements to meet the business goals over the specified business planning period.
- Asset management strategies and initiatives are specific and enable the asset capability to be approved and delivered.
- A risk-based approach has been used to determine the best submissions to approve.
- Asset management plans developed based on the demand requirements for the asset and the level of service and responsibilities to achieve those requirements.
- Asset management plans includes the operating context, life cycle costing, maintenance and operating plans.
- Asset management plans are reviewed to take account of ongoing issues and risks.

Key Requirements:

3.1 – Performing assessments on existing critical assets to confirm they are fit-for-purpose and able to achieve asset management goals:

- **Assessing** asset **condition and integrity** to determine replacement, upgrade, repairs or rebuild requirements.
- **Using lifecycle** cost **analysis** to determine economical end of life and replacement plans.
- **Using asset capability to meet** current, future or alternative usage **requirements**.
- **Recognising** the impacts of l**egislative, standards and guidelines** changes.
- **Developing options to retire assets** (sell, scrap, use for spare parts, recommission, closure planning).

3.2 – **Developing** and assessing asset **management plans** and initiatives to confirm their effectiveness to achieve the business plan.

- **Developing supporting plans** for each initiative.
- Considering, for each asset management plan or initiative:
    - Critical Business Risks.
    - Take up of new technologies.
    - New assets.
    - Improvement to existing assets.
    - Changes to business systems and processes.
    - Improvements in supplier arrangements.
    - Changes in organisational capability.
    - Existing plans assessed in line with new initiatives.
    - Changes in operating envelope or conditions.

3. Asset Management Planning (continued)

3.3 – Establishing the capability to manage the workload to deliver the asset management plans:

- **Engaging the resources and skills** required **to deliver the plans**.
- Developing the **organisational capacity and skills** to manage projects and shutdowns.
- Planning to coordinate and deliver on all initiatives and projects.
- Identifying needed external support.

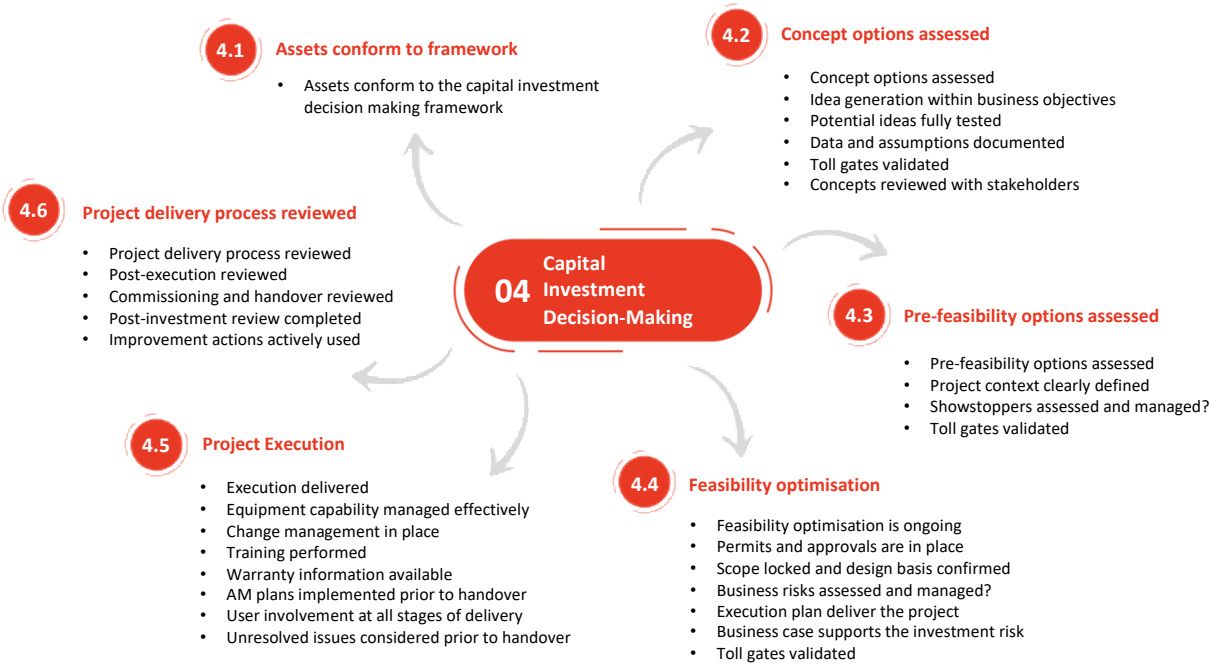
3.4 – **Reviewing** asset management **plans regularly** between budget periods and re-forecasting for changes in operational plans, new issues and risks:

- Establishing a review period.
- Ensuring progress against all initiatives and plans is available.
- Considering the effects of new issues or risks.
- **Communicating changes** to initiatives and plans.
- **Controlling plan changes** through change management processes.

4. Capital Investment Decision-Making

Intent

Capital and major operating investment decisions deliver new or modified assets on time, in budget, and have the fit-for-purpose capability to meet the business objectives.



Drivers:

- Achieving the business goals and objectives.
- Optimising the lifecycle costing and operational performance by ensuring the purchasing or upgrading of assets are fit for purpose, cost effective, safe and efficient to operate and maintain and are well supported within the supply chain.
- Ensuring that the required capability is delivered by involving stakeholders through a careful assessment of the options, their associated functional requirements and risks.
- Understanding that the application of Glencore Copper's Capital Management Framework is required for capital investment decision-making when certain thresholds are met.

4. Capital Investment Decision-Making (continued)

Outputs:

- Forecasted business case objectives are achieved for new or modified assets.
- The required asset is presented ready for commissioning, fit for purpose, on time, and within budget.
- All asset management plans and the associate supply chain requirements are in place to support the asset over its whole life cycle prior to handover.
- Risks have been identified and suitable controls implemented.
- The capability limits including Standard of Operation (SOP) of the asset are documented and communicated to operations.
- Success and learnings of each project are documented available for future projects.

Key Requirements

4.1 – Delivering new or modified **assets** that conform to **Glencore Copper's Capital Management Framework**, purchasing and supply requirements as well as specific asset management key requirements.

4.2 – Progressing ideas or opportunities in the concept phase with the intent to achieve the strategic plan and objectives:

- **Idea generation** in context **with the business objectives**.
- **Testing** the potential of **ideas** with stakeholders through high level analysis using benchmarking, risk analysis and financial modelling.
- Developing supporting data and assumptions that advance business case (**Toll Gate**).

4.3 — Generating and **reviewing pre-feasibility options** to achieve business requirements and selecting the preferred alternatives for further analysis:

- **Clearly defining the project context** as to achieve the business objectives.
- Potential **showstoppers** have been adequately **assessed and managed**.

4.4 — Optimising the preferred alternatives and selecting the best alternative or option to proceed to the 'execution' stage:

- All permits and approvals are in place.
- The scope is locked and the design basis is confirmed.
- Business risks have been adequately assessed and managed.
- The execution plan will deliver the project on time, in budget and at the required specification to meet the business objectives.
- The business case supports the investment risk.

4.5 – **Executing** new or modified asset **deliverables** prior to handover to operations:

- **Validation of equipment capability.**
- Engineering changes, statutory compliance documentation, as-built drawings, Bill of Materials (BOMs), critical spares, tooling, training requirements and safe work procedures.
- Operational and Engineering **change management** processes in **place**.
- **Training** being **performed**.
- All **warranty information** being **available**.
- The asset management **strategy** being **implemented prior to handover**.

4. Capital Investment Decision-Making (continued)

- Evidence of **user involvement at all stages of** asset **delivery**.
- **Issues not resolved** before the end of commissioning are **formally handed** over or the project remains in execution until resolved.

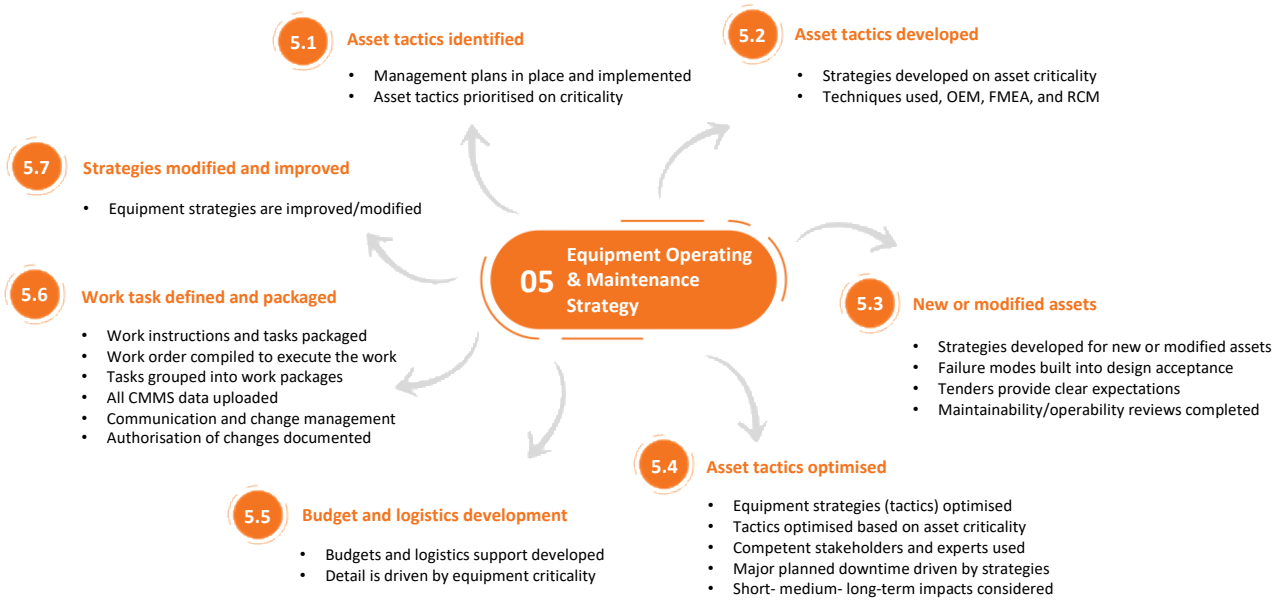
4.6 – **Reviewing** new or modified asset performance and **the project delivery process**:

- Completing a **post-execution review**.
- Successful **commissioning and handover**.
- Completing a post-investment **review**.
- Establishing and **actively using** a capable process for implementing project execution and business performance **improvement actions**.

5. Equipment Operating and Maintenance Strategy

Intent

Assets have management processes and activities in place, determining the operating and maintenance strategies to support the Asset Management objectives and goals.



Drivers:

- Recognise that business objectives and goals continually challenge demands on assets that directly correlate with an asset's operating and maintenance strategy.
- Documenting decisions on asset operation and maintenance due to demand changes ensures a better understanding of the influence asset constraints and lifecycle have on operating and maintenance resources, energy and water usage, process consumables and spares inventory.
- Implementing risk mitigation within the strategy that addresses critical assets (or systems) bottlenecks and constraints.
- Developing Operating Equipment Manufacturer (OEM) recommendations, after considering all potential operating contexts and whether it is fully appropriate for application at a specific operation.
- Determining an asset's potential performance mostly during the design phase of its lifecycle. Application of Reliability Centred Maintenance (RCM) Principles in design acceptance processes prevents inadequate designs reaching operations.

5. Equipment Operating and Maintenance Strategy (continued)

- Recognising equipment strategy, given its demonstrated impact on business performance, in all external asset management frameworks.
- Developing and sharing effective asset management strategy improvements, rely on documentation of strategy selection, its implementation and timing.

Outputs:

- An Operation and Maintenance strategy that is based on a documented operating context.
- Asset (systems) criticality determined by applying a recognised risk assessment process. Controls shall be applied to high criticality assets that reduces risk to an acceptable level.
- Analysis of Equipment Failure modes, their effects and application of a hierarchy of controls during the design acceptance for new or upgraded assets.
- Maintenance directed at ensuring equipment remains fit for purpose over its whole life cycle. Risks from equipment failures are reduced to tolerable levels.
- Changes to the asset strategy are documented, approved and the reasons captured. All supporting logistics (Labour, Spares, Work instructions, replacement capital, energy, water, process consumables) are updated accordingly.
- The achievable performance of the asset, given the current design and operating context, is forecast against the current and future demand requirements.

Key Requirements

5.1 – Establishing **management plans** to identify when equipment operating and maintenance strategies are **developed** and reviewed to meet business objectives.

5.2 – **Developing** equipment operating and maintenance **strategies based on** asset **criticality** utilising adequate procedures, tools, training and standards.

5.3 – **Delivering** equipment operating and maintenance **strategies for new or modified assets**:

- Identifying **failure modes** and the effects on asset functions and building **into design acceptance** processes.
- Ensuring **tenders provide clear** descriptions of expected life and functional **requirements**.
- **Reviewing maintainability and operability reviews**, including the requirements defined during functional safety reviews.
- Reviewing asset monitoring, warning and shutdown capabilities.
- Change management implemented.

5. Equipment Operating and Maintenance Strategy (continued)

5.4 – Developing and **optimising equipment** operating and maintenance **strategies** to meet present and future asset demands:

- Defining clear failure modes, effects and criticality analysis for each level of equipment, based on its forecast whole-of-life operating context and criticality to the operation.
- Considering and selecting operating and maintenance tasks by:
    - Using RCM decision-making principles.
    - Referencing OEM recommendations and current tasks for similar equipment.
    - Using requirements under legislation and company guidelines.
    - Understanding available leading practice condition monitoring techniques.
    - Including the current and forecast operating context.
- **Using competent** teams with a cross-section of **stakeholders and experts**.
- Constructing timelines for **major planned downtimes required by** the maintenance **strategy**.
- **Understanding the short-, medium- and long-term impacts** on asset performance due to unplanned deviations from the maintenance and/or operating plans.

5.5 – **Defining budgets and** the required **logistics** to **support** the operation and maintenance strategy, using a level of **detail based on** equipment **criticality**:

- Forecasting performance, resources and costs (CAPEX/OPEX).
- Utilising risk mitigation and initiatives process.
- Considering consumables, energy and water requirements.
- Recognising spares and services required (when, where and under what agreements).
- Defining corrective work to recover or repair failed equipment.
- Identifying the specialist tools and equipment needed to operate and/or conduct maintenance on the asset.
- Using fault-finding tools and systems to aid timely identification and the repair of equipment failures.

5.6 – Defining, grouping and **packaging tasks or work instructions** to minimise downtime and maximise resource utilisation:

- Ensuring **work** order **compilation** and registration is to a documented standard.
- **Grouping tasks into work packages** to best align with operating patterns and minimise planned downtime.
- **Uploading** of **all** Computerised Maintenance Management System (**CMMS**) **data**.
- Committing to **communication and change management**.
- **Documenting** the **authorisation of changes**.

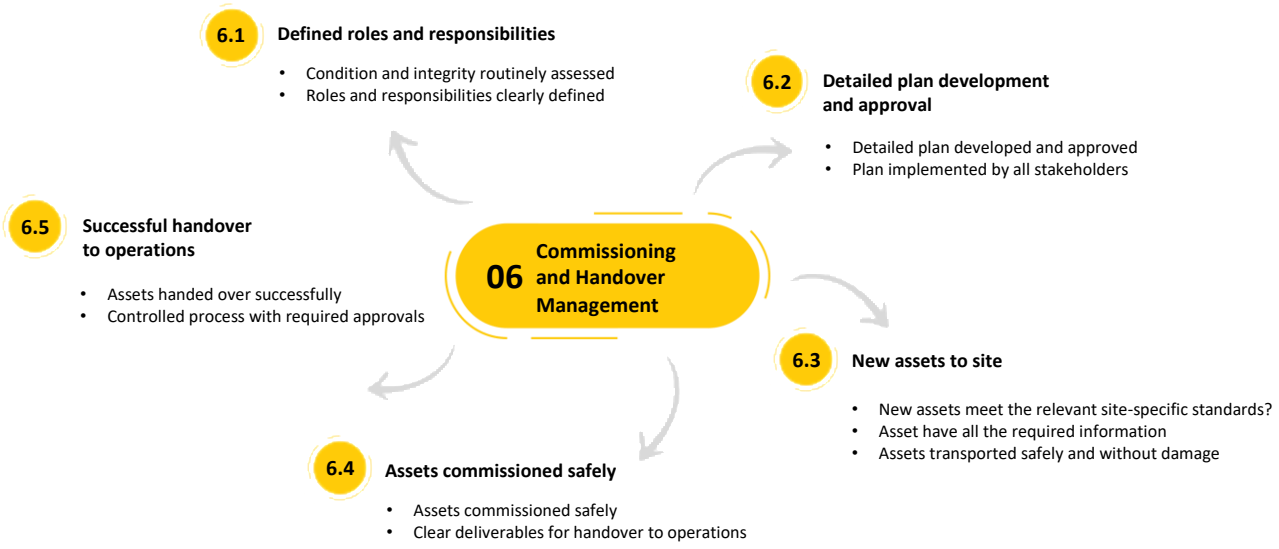
5.7 – **Improving, modifying and documenting equipment** operating and maintenance **strategies** when there are changes to operational context or demand:

- Setting review periods for implementation, based on equipment criticality.
- Applying changes in legislative requirements and engineering practice standards.
- Receiving alerts or improvements.
- Investigating identified flaws in the asset management strategy.
- Incorporating significant changes when the operating context occur.

6. Commissioning and Handover Management

Intent

New or modified assets are managed through delivery, installation and commissioning to confirm they are fit-for-purpose and demonstrate capability to meet their functional requirements before handover to operations.



Drivers:

- Commissioning and handover of new or modified assets are a critical stage in controlling HSEC risks especially when the asset has never been operated or maintained at site before.
- Returning assets to service after maintenance or storage undergo the same formal processes as the introduction of new assets.
- Commissioning confirms the asset will perform all of its required functions.
- Ensuring a controlled ramp-up and phased handover of an asset to operations allows time for training and competence to be put into practice.
- Identifying and controlling most quality (materials and workmanship) and installation quality issues during in the commissioning process.
- Applying of Glencore Copper's Capital Management Framework for the commissioning and handover for assets that meet the investment and or risk threshold.

6. Commissioning and Handover Management (continued)

Outputs:

- The required assets are introduced into service, safely and meet the operational performance targets.
- Clear plans exist for the commissioning phases and the handover of control of the asset.
- All asset management plans, supporting logistics and information are in place prior to handover.
- Commissioning records are created for future reference.
- Operations formally accept the responsibility for the asset once commissioned.
- Risks and issues have been identified and suitable controls and action plans implemented.
- Updated drawings, documentation, training programs, maintenance programs to record any modifications made during commissioning.

Key Requirements

6.1 – **Defining** and communicating clear **roles and responsibilities** for each phase, from introduction to site on to handover to operations:

- Authority to introduce asset.
- Responsibilities for transport, installation and commissioning.
- Authority to approve the completion of installation and commissioning phases including break in or out of scope work.
- Responsibilities and authorities (including support teams) during hand over to operations.

6.2 – **Developing a detailed plan approved by all stakeholders** prior to commissioning of new or refurbished assets:

- Testing and acceptance of plant or equipment.
- Assessing all risks and mitigation strategies.
- Detailing a project plan with timings, resources, operational involvement and interfaces.
- Implementing a pre-commissioning inspection defect list and rectification program.
- Completing an asset forecast (including ramp-up) within the site operating plan.

6.3 – **Introducing assets** to site will be required to:

- **Meet the relevant site-specific standards**.
- Have **all** the **required asset information**.
- Be **transported safely and without damage**.

6.4 – **Commissioning all assets safely** with **clear deliverables for handover to operations**:

- Aligning the commissioning plan and site Health and Safety (H&S) Management plans.
- Communicating start, progress and handover.
- Completing the required asset performance criteria prior to handover.
- Incorporating operational requirements during dry and wet commissioning.
- Involving all personnel adequately in commissioning.
- Providing adequate training and information prior to handover.

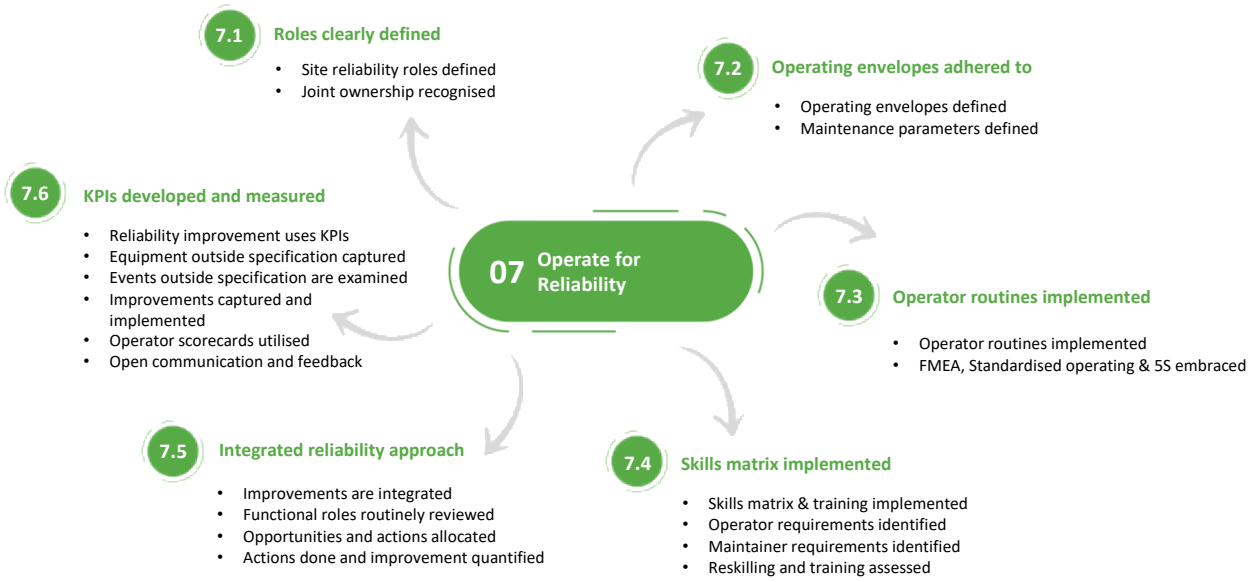
6.5 – **Handing over assets** to operations **successfully** and safely conducted under a **controlled process and all required approvals** from the authorised asset owner:

- Updating CMMS, cataloguing and warehousing of spares, removal of obsolete spares, training information and budget adjustments.
- Updating all asset information with any changes from installation and commissioning.
- Handing over to operations and maintenance all outstanding issues and responsibilities agreed for completion.

7. Operate for Reliability

Intent

Operational influences on equipment reliability are well understood and managed to sustain asset integrity and performance.



Drivers:

- Asset reliability is not the sole domain of maintainers.
- How equipment is operated has a significant impact on its reliability. Correct set-up, operating within specifications, high standards of housekeeping / cleaning and reporting of defects are behaviours operators can apply to improve asset safety and reliability.
- Operate for reliability is based on the belief people will do the right thing if they understand what to do and why it is important.
- Different types of equipment and their application requires specifically tailored awareness programs that ensure they are operated appropriately.

7. Operate for Reliability (continued)

Outputs:

- An integrated approach to reliability improvements that incorporates operators, maintainers and technical functions.
- Improved safety outcomes driven by increased awareness, a clean work environment, better equipment condition and more rapid fault identification and rectification.
- Enhanced operational performance (improved equipment reliability, reduced wastage and lower costs) achieved by equipment operating within their design parameters.
- Behavioural change – people do the right thing through awareness and feedback.
- Standard work procedures consistently applied across all shifts.
- An up to date Skills Matrix and Training Plan.
- Equipment damage events investigated to identify and address root causes.
- Training material developed and applied to existing employees and contractors and for new starters including training in basic inspections and maintenance.
- Operating envelope compliance measured and analysed to minimise risks to asset health and integrity.

Key Requirements

7.1 – **Defining** clearly and communicating regularly the r**oles** and responsibilities of the technical teams, operators and maintainers:

- Outlining their responsibilities.
- Detailing the interfaces between departments and functions.
- Using escalation process and authorities as the health status of equipment changes.
- **Recognising** the importance of **joint ownership** between operations, maintenance and technical functions.

7.2 – Communicating and **adhering to** assets' design **operating envelope and maintenance parameters** by all departments:

- Deploying operating specifications:
    - Assessing demand requirements to meet forecast production rates.
    - Respecting the limits of asset design specifications.
    - Recognising environmental factors.
    - Identifying acceptable operational tolerances.
- Deploying maintenance specifications:
    - Capturing the influence of variable demand requirements on maintenance activities.
    - Using design specifications and regulatory requirements.
    - Establishing quality control for maintenance activities.

7.3 – Developing, communicating and **implementing operator routines**:

- **Embracing** the failure modes effects analysis methodology.
- Introducing a Standardised operating philosophy and work practises.
- Empower operator condition monitoring through observations utilising their senses and the equipment information systems.
- Maintained housekeeping at **a defined standard such as 5S** (Sort, Set in order, Shine, Standardise and Sustain).
- Implementing a trigger action response plan (TARP) to manage unplanned events.

7. Operate for Reliability (continued)

7.4 – Developing and **implementing** a detailed **skills matrix and training** plan to underpin improving asset performance and safety:

- **Identifying operator** skills and training **requirements**.
- **Identifying maintainer** skills and training **requirements**.
- **Measuring** current capabilities and trigger requirements for **reskilling and** or **retraining**.

7.5 – Developing and implementing an **integrated** approach to reliability **improvements** that incorporates operators, maintainers and technical functions:

- **Routinely review** the current performance of the asset(s) and the **roles** that each function is in control of.
- **Identify opportunities and allocate actions** and responsibilities.
- Review previous **actions for implementation** and quantify **improvement.**

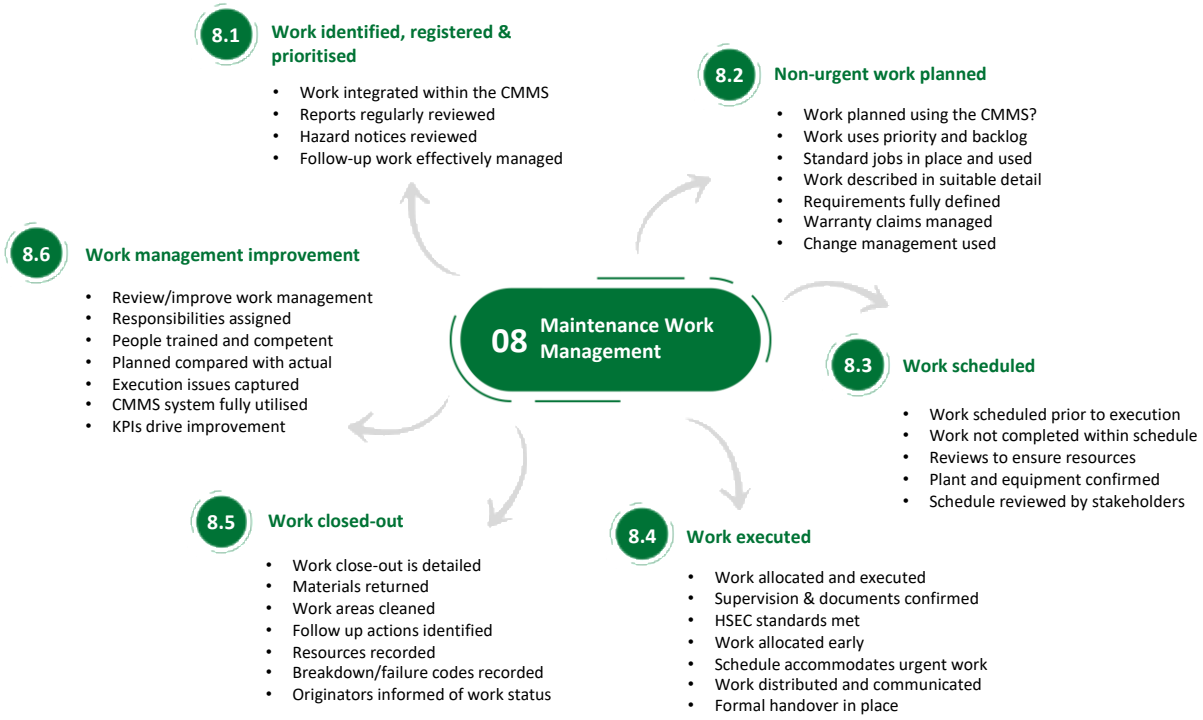
7.6 – Developing and implementing a series of operator KPIs (and their supporting metrics) to monitor asset performance improvement:

- **Capturing** the **events of equipment** operating **outside specification**.
- Examining any trending and analysis of events and associated connections to equipment failure.
- **Capturing and implementation of improvement initiatives**.
- **Using operator scorecards**.
- Adopting an **open communication and feedback** process.

8. Maintenance Work Management

Intent

All Maintenance work is identified, prioritised, planned and progressed to completion in an efficient and timely manner and the overall completion rate does not pose a risk to people, equipment, the environment or achievement of the business goals.



Drivers:

- Managing health and safety risks in maintenance activities primarily by maintenance work management processes and procedures.
- Recognising that maintenance is the key process within asset management, consumes significant resources and time. Consequently, it should be defined, understood and managed in detail.
- Coordinating people, parts, procedures and tools to provide an environment where maintenance is performed efficiently and effectively.

8. Maintenance Work Management (continued)

- Minimising the consequences of equipment failure modes via the timely execution of maintenance strategies.
- Understanding that asset availability is dependent on the efficient use of planned maintenance. Several aspects of maintenance work management affect this including:
    - Coordinating operations and maintenance during planned downtimes.
    - Using opportunities to complete well-prepared work minimises wasted time, reduces personal risk.
    - Avoiding significant wasted effort trying to improve the plant and equipment strategy when the root cause is inadequate maintenance work management.

Outputs:

- Safe execution of maintenance activities.
- Maintenance work is progressed based on criticality.
- A short-term forecast (schedule) exists for plant/equipment, resources and spares.
- Efficient utilisation of resources within the planned operational downtime.
- KPIs exist for the Maintenance Workflow.
- Increased equipment availability and reliability.
- Reduced cost from unplanned work.
- Risk of asset failures are kept within tolerable levels.
- The ability to measure the effectiveness and improvement of the maintenance strategy and the logistics (spares, tools, machines etc.) that are in place to support the strategy.
- Ability to accurately forecast costs (OPEX/CAPEX), resources and capability required to manage Maintenance Work Management.

Key Requirements

8.1 – **Identifying, registering and prioritising** by risk all maintenance **work** in the CMMS:

- **Reviewing regularly** all **reports** (trade, operator, shift, supervisor, condition monitoring, improvement, preventative and predictive tasks etc.) along with defect management systems (out-of-service tag, defect notices, conveyor roller tags etc.) and identifying and registering any outstanding work.
- **Reviewing** all internal and external **hazard notices** (including statutory requirements) and identifying site-relevant maintenance actions.
- **Identifying**, reviewing and prioritising any **follow-up work** from previously executed maintenance.

8.2 – Planning all non-urgent work prior to execution with a level of detail based on the risk in delaying and during execution:

- **Utilising CMMS work orders**.
- Planning all **work** is in **priority** order, including updating **backlog** priority.
- Backlog reviews with stakeholders.
- Creating and **using standard job** templates for repetitive work.
- **Describing the work** (Standard Work Instructions (SWIs), Job Safety Analysis (JSAs), risk assessments) **in suitable detail** to enable safe and efficient execution.
- **Defining** and arranging resource and inventory **requirements**.
- **Identifying** work under warranty and initiating **warranty claims**.
- Ensuring work requiring modification **complies with change management** protocols.

8. Maintenance Work Management (continued)

8.3 – **Scheduling** maintenance **work ready to be executed**, for any given period, in priority order:

- Post-reviewing the equipment criticality ranking, preventive maintenance due, planned work backlog, and **previously scheduled work not done** but still required.
- Reviewing resource availability for the schedule period, including leave, non-maintenance work (training, toolbox talks, etc.) contractor availability, allowance for breakdown work, operational requirements and roster patterns.
- **Confirming plant and equipment** location and availability.
- Confirming via **scheduling review** meeting **with** all key **stakeholders** and delivering a work plan highly likely to succeed.

8.4 – Allocating, **supervising and executing** maintenance **work** safely, efficiently and to quality specification:

- Ensuring no work is executed until **HSEC standards** for the work are **met**.
- **Allocating work** to allow adequate information for safe, efficient and quality **execution**.
- Accommodating breakdown work in the schedule and carrying out the resulting opportunity maintenance.
- Ensuring short-cycle re-scheduling of work occurs based on progress.
- Ensuring **work** is efficiently **distributed and communicated** between shifts.
- **Confirming** that **formal handover** to operations has taken place.

8.5 – **Closing off** in a timely manner all **work** completed or deemed unable to be completed in the current scheduling period:

- Returning all unused materials, tools, equipment, warranty and repairable items.
- Cleaning up the work area.
- Documenting in suitable detail what was done or found for all follow up actions to be identified and easily planned.
- **Recording of all resources** used to complete the work including contractors, internal labour and equipment.
- Capturing actual work done and reasons for why work was required, including breakdown work.
- Identifying and recording any further work required.
- **Recording failure** causes.
- **Informing originators** that **the work is done**.

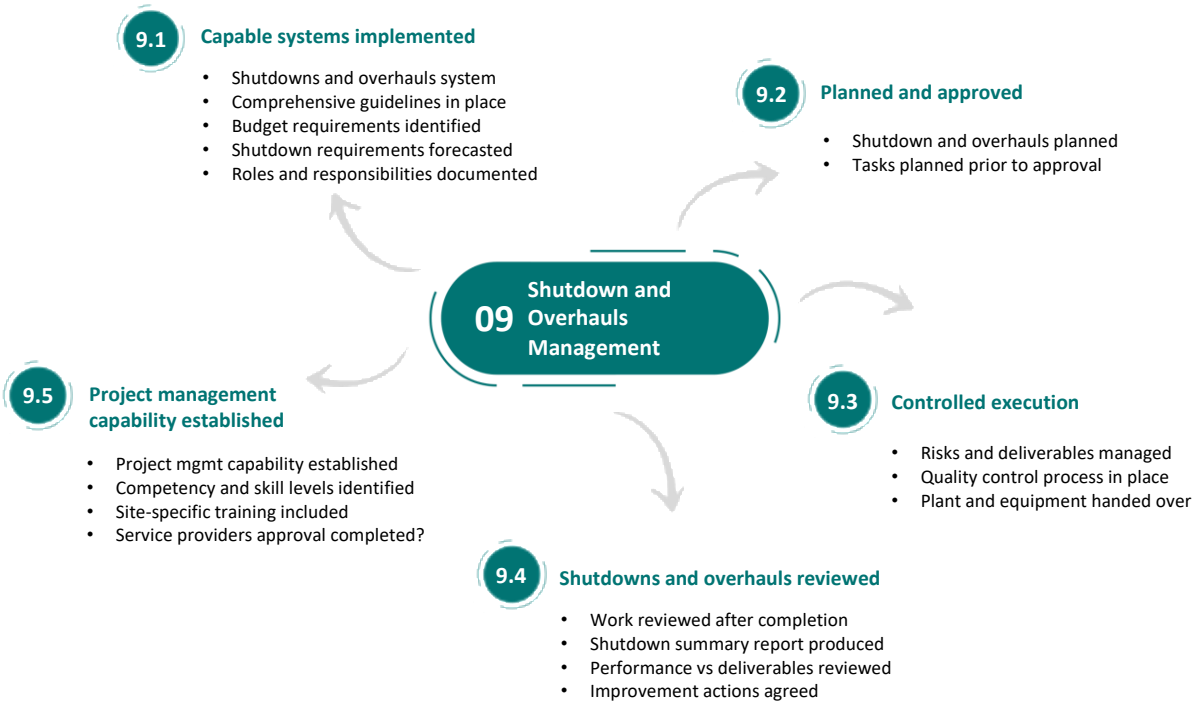
8.6 – Establish a management structure capable of **analysis and improvement** of maintenance process efficiency and associated outputs:

- Documenting of workflows for maintenance with assigned responsibilities and accountabilities.
- Ensuring **personnel** involved in maintenance **work management** are adequately **trained and competent**.
- Recording work completion statistics for comparison with planned estimates.
- **Reviewing of actual** completion **against plan** for the period.
- **Capturing** and actioning **execution issues**.
- **Utilising the CMMS system** capability by analysing data to improve asset reliability and maintenance performance.
- **Establishing** suitable and capable **KPIs** (and their metrics) **and continuously implemented actions** for efficient maintenance delivery.

9. Shutdown and Overhauls Management

Intent

Processes for shutdowns and equipment overhauls have management systems in place for plant or equipment to be removed from service and subsequently returned safely, on time, within budget and fit-for-purpose.



Drivers:

- Recognising that compared with day-to-day maintenance, shutdowns occur over longer time periods, contain interaction between multiple tasks and have significant costs. Additional project management competencies and processes are in place to achieve successful outcomes.
- Implementing a shutdown and overhaul strategy as an outcome of the business planning process and as a requirement for assets achieving business goals and objectives.
- Recognising the pre-planning and the associated procurement actions needed for long lead-times on materials and the required specialist skills in advance of the execution phase.
- Involving all stakeholders in the shutdown/overhaul planning process to identify and agree on the scope.

9. Shutdown and Overhauls Management (continued)

- Increasing personnel numbers and non-standard tasks within the work environment require detailed risk assessments to monitor that HSEC and project risks are at an acceptable level.
- Understanding that detailed planning and scheduling limits shutdown and overhaul 'scope creep' can impact unfavourably on costs and forecast production.
- Recognising that the timely plant handover by production to the shutdown or overhaul team is critical in optimising shutdown performance. Similarly, quality control of major completed maintenance and recommissioning of equipment or plant is crucial to ensuring a 'fit' to return to service.
- Identifying that overhauls and shutdowns are opportunities to modify plant and equipment and improve asset operation, cost and safety performance.
- Applying Glencore Copper's Capital Management Framework is required on meeting certain investment and or risk thresholds.

Outputs:

- Clear identification of the shutdown activities, leadership structure and accountabilities are communicated and agreed with all stakeholders.
- Procurement and contract processes in place to achieve project timelines.
- Shutdowns and Overhauls start and end in the agreed timeframes.
- A recommissioning plan is in place that incorporates a formal handover to operations, quality checks and tests to confirm the plant or equipment is fit to return to service.
- HSEC risks levels are managed to acceptable levels.
- Shutdowns and Overhauls deliver the agreed outcomes.
- Shutdown and Overhauls management capability is in place.
- Shutdown/Overhaul review and asset information is updated.

Key Requirements

9.1 – Developing and implementing effective **shutdown and overhaul** management **systems**:

- Establishing a site shutdown and overhauls management process.
- **Using guidelines** for planning, handover, execution and hand back.
- Referencing cost control guidelines to achieve **budget requirements**.
- **Forecasting** the site **shutdown** and overhaul **requirements**.
- **Describing the roles, responsibilities** and authorities for management of shutdowns and overhauls.

9.2 – **Planning shutdowns and overhauls prior to** execution **approval**:

- Defining handover duration and operational requirements and hand back by the project teams are and inclusion in the schedule.
- Meeting all requirements of maintenance work planning.
- Resolving capability issues and solutions within the scopes.
- Conducting risk assessments for key stages in shutdown.
- Creating work breakdown structures and timelines.
- Ensuring regular stakeholder engagement and planning progress reviews.
- Understanding that project budgets and cost control measures allow earned value reporting and cost grouping from work orders.
- Establishing project management and supervision requirements for contracts.
- Developing project specific management plans including, HSEC, quality, traffic, logistics, etc., as necessary.

9. Shutdown and Overhauls Management (continued)

9.3 – Executing shutdowns and overhauls in a controlled manner to ensure identified **risks** are **managed and deliverables** are **met**:

- **Handing plant or equipment over** to the project team at the time and state as committed by the operations.
- Meeting all maintenance work execution requirements.
- Nominating a project manager/shutdown coordinator.
- Communicating project management plans and monitoring compliance.
- **Ensuring supervision controls** cover all shutdown personnel.
- Updating and re-forecasting the project plan as required.
- Confirming that the redistribution of updated work packages is carried out.
- Confirming that re-commissioning and handover approval back to operations have taken place.
- **Implement quality control process** for critical asset activities

9.4 – **Reviewing completed shutdowns and overhauls** to determine success and potential improvements for future shutdowns:

- **Producing a shutdown summary report**.
- **Reviewing performance against** defined **deliverables**.
- **Agreeing to improvement actions** for future shutdowns logged.

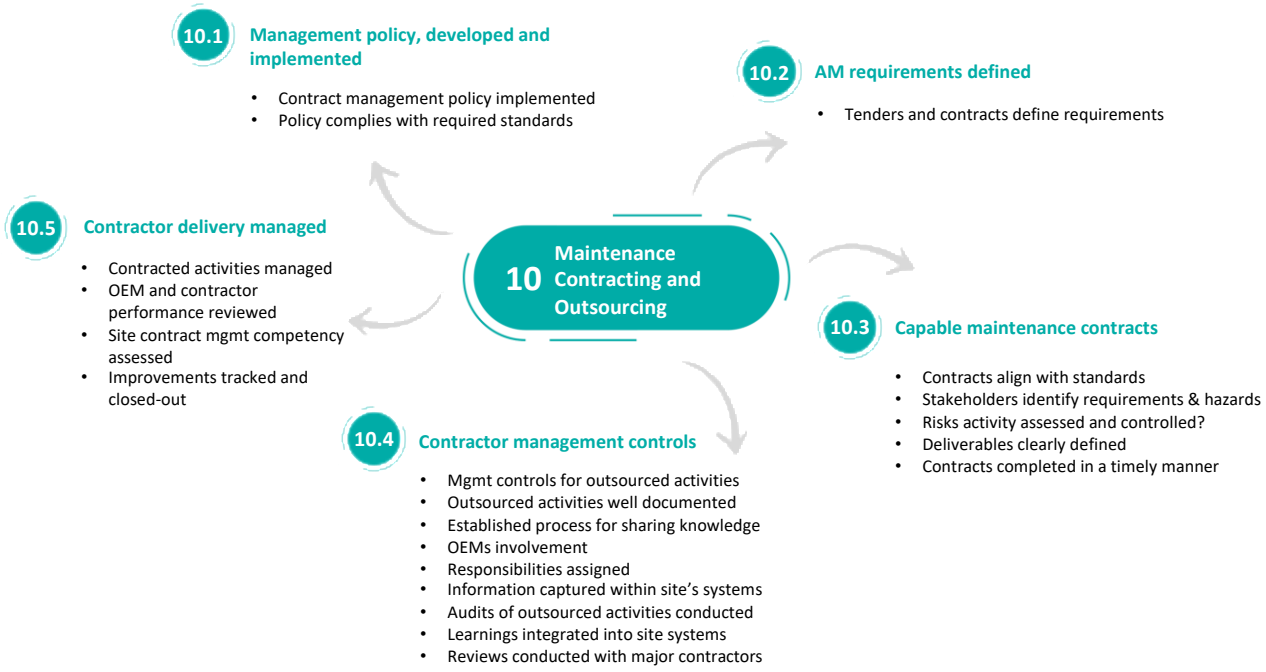
9.5 – **Establishing**, reviewing and implementing the required shutdown and overhauls **project management capability** available to site:

- **Defining competency and skill levels**.
- **Including site-specific training** requirements.
- Identifying and **approving service providers** and their employees.

10. Maintenance Contracting and Outsourcing

Intent

Contracted and outsourced maintenance activities are conducted in line with the Asset Management Framework requirements and support delivery of Asset Management objectives.



Drivers:

- Retaining Glencore Copper accountability for contracted asset management activities and effective controls are required to manage risks.
- Controlling contractor competence to perform work safely and to a suitable quality through the contract process (contractor selection through to execution).
- Recognising the opportunity in longer-term contracts to deliver performance improvements and control resource consistency.
- Appreciating OEMs have significant influence on the delivery rate of product and available technology improvements with proactive involvement this may be directed towards Glencore's priorities.
- Understanding that the performance of outsourced activities affects an asset's cost-effectiveness, process efficiency levels, resourcing levels and team morale.
- Managing the key issues in contractor competence and parts supply.
- Extending asset management governance to outsourced activities to align with Glencore Copper objectives.

10. Maintenance Contracting and Outsourcing (continued)

- Adding significant value to performance-based contracts by allowing for adjustments in goals and contract terms for changes to operational goals and objectives.

Outputs:

- Flexibility in choice of supplier or contractor.
- Risk based approach for selection of non-OEM sourcing for critical spares.
- Activities carried out by contractors, suppliers and OEMs align with short and long-term business objectives.
- Ongoing control of Asset Management activities and access to information.
- Asset Management risks for contracted activities are controlled.
- A viable alternative to owner maintained is available where required.
- Strategies and performance targets are in place to manage OEM interactions.
- Issues are resolved within an acceptable time and solutions are effective.
- Roles and responsibilities are in place to manage critical business interactions.

Key Requirements

10.1 – Developing and **implementing a contract management policy** incorporating maintenance, OEMs and outsourced suppliers at each site:

- **Complying with** HSEC **standards**, shutdown and overhaul management and the Glencore Copper's procurement and supply governance.
- Establishing asset management selection criteria for preferred contractors by:
    - Confirming safety performance and systems.
    - Complying with asset management systems.
    - Exhibiting asset management competencies.
    - Showing a capacity to understand and support asset lifecycle requirements.

10.2 – **Defining** clearly the asset management **requirements in tenders and contracts**:

- Describing scope, boundaries, activities and deliverables.
- Establishing KPIs for measuring contractor performance.
- Identifying risk assessments and acceptable assessment methods to be used by the contractor.
- Implementing a process for ongoing control of outsourced asset management activities.
- Complying with the asset management framework requirements.
- Agreeing on how ongoing scope changes are carried out and cost allocation.
- Establishing a mechanism for enabling ongoing changes to contracted arrangements.
- Establishing mechanisms to identify and renegotiate contracts to match operational target changes.

10. Maintenance Contracting and Outsourcing (continued)

10.3 – Developing and implementing **capable maintenance contracts aligned with** Glencore Copper's contracting **standards** and site contract processes:

- Consulting with **stakeholders to identify requirements and** potential **hazards**.
- **Assessing** contractual **controls of risk** associated with contracting out maintenance activity are assessed by:
    - Identifying HSEC risks associated with **activity**.
    - Agreeing on the performance of contracted activity.
- **Defining clearly deliverables** such as:
    - Ongoing standards of service required.
    - Ongoing standards of documentation and reporting.
    - Accessing of documentation and reporting.
    - Establishing supervision requirements.

10.4 – Implementing **management controls for outsourced activities** to confirm asset management requirements are always met:

- **Documenting** the scope of **outsourced activities** and interfaces with site.
- **Establishing a** clearly defined and communicated **process for sharing knowledge** between the site and the contracted service provider.
- Ensuring **involvement with OEMs** influencing future design and modifications and enabling adequate support for existing equipment.
- **Assigning** site **responsibilities** for managing outsourced activities.
- Managing contractor collected and generated **information in** a system integrated with **the site's systems**.
- **Conducting** regular reviews and **audits of outsourced activities** against Asset Management Framework requirements.
- **Integrating learnings** and improvements from outsourced activities into site and Glencore Copper.

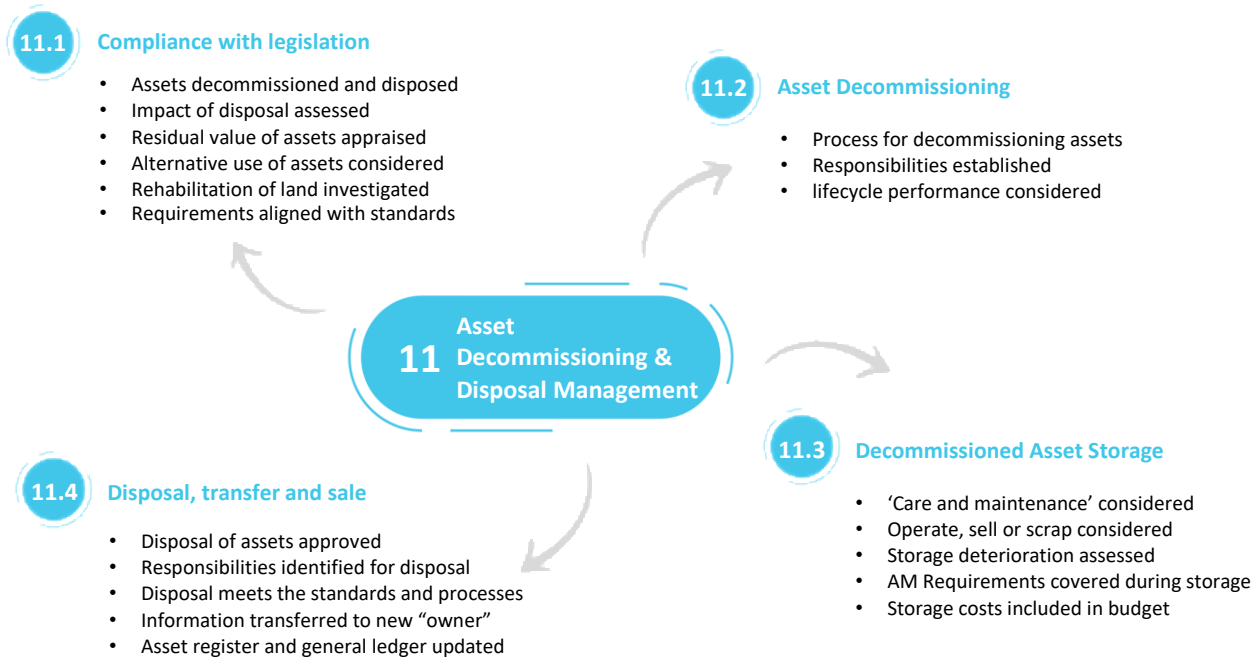
10.5 – **Managing** all **contracting** to confirm delivery of asset management objectives for contracted and outsourced **activities**:

- **Reviewing OEM and contract performance** regularly to identify and implement improvements.
- **Assessing** gaps in **site contract management competency** and development plans.
- **Tracking issues** and progress rectification.

11. Asset Decommissioning and Disposal Management

Intent

Processes ensure compliance with internal and regulatory requirements for the decommissioning and disposal of assets.



Drivers:

- Understanding that Glencore Copper, as a large group, can transfer assets from one site to another. Managing this transfer process to maintain asset integrity, commercial requirements and legal obligations.
- Meeting legislative obligations for the decommissioning and disposal of assets.
- Amending all systems supporting management of assets in operation to reflect asset decommission decisions.
- Viewing the decision to decommission assets as an opportunity to review lifecycle performance.
- Meeting Glencore Copper procedures for asset decommissioning and disposal.

11. Asset Decommissioning and Disposal Management (continued)

Outputs:

- Clear authorities and responsibilities for decommissioning and disposal of assets.
- Asset Management systems show the correct status of decommissioned assets.
- Commercial requirements managed for disposal of assets including sale, transfer or discarding of assets.
- Legal obligations managed for disposal of assets.
- Asset Integrity managed for stored assets.
- Health, Safety, Environment and Community risks are well managed.

Key Requirements

11.1 – Developing processes for **assets** that address legislative requirements for the **decommissioning and/or disposal** of individual assets through to site/mine closure:

- Assessing the environmental impact of disposal, including hazardous waste.
- **Appraising the residual value of** the **assets**.
- **Considering alternative uses for** decommissioned **assets**.
- **Investigating land rehabilitation** including decontamination.
- **Aligning work with** Glencore Copper's Mine/Site Closure Planning **requirements** from the Environmental Policy.

11.2 – Developing and implementing at site a **process for decommissioning assets**:

- **Establishing** authorities, **responsibilities** and timelines for decommissioning assets.
- Ensuring a process for decommissioning assets covers at least:
    - Updating the CMMS status for assets.
    - Updating of Planned Maintenance schedule status.
    - Managing Inventory.
    - Updating of commercial status.
    - Terminating of contracts.
    - Assessing risks associated with decommissioning.
    - Ensuring competency and safe work procedures.
    - Updating the Asset Register.
- **Analysing lifecycle performance** for use in future decision-making.

11.3 – **Developing and implementing arrangements for** the storage and **'care and maintenance'** of decommissioned assets to ensure tolerable risk levels for HSEC and any future recommissioning requirements:

- **Planning** for future requirements as **to operation, sale or scrap**.
- **Assessing** risks from plant and equipment **deterioration during storage**.
- **Covering maintenance requirements during storage**.
- **Budgeting costs for storage**.

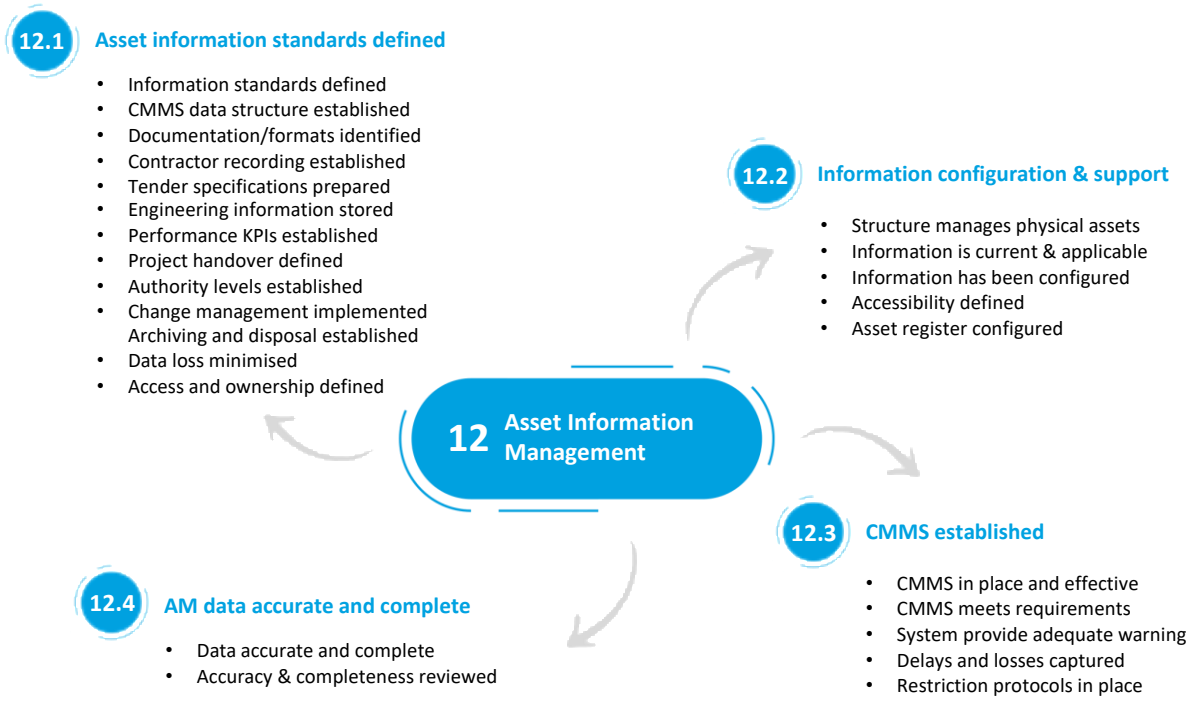
11.4 – Ensuring the **disposal**, transfer or sale **of assets** is **approved** by an authorised person and conducted in line with a defined process, which confirms compliance with all Glencore Copper corporate and legislative requirements:

- Identify authorities and responsibilities defined for the disposal, transfer or sale of assets.
- Ensure the **disposal** of assets **meets** the required **legislative and** internal **processes**.
- Ensure the transfer of records and information required under legislation.
- **Updating the asset register and general ledger**.

12. Asset Information Management

Intent

Information is identified for managing assets through their lifecycles and an effective information and document management system is implemented and maintained.



Drivers:

- Collecting, managing, reporting and overall governance of asset information to support effective delivery of the asset management strategy and objectives.
- Documenting to ensure consistent task quality by maintenance and operating procedures.
- Understanding the availability of accurate and current information cuts planning time and reduces downtime.
- Ensuring correct up-to-date spares information to improve repair intervals and potentially reduce hazards.
- Ensuring correctly registered assets in the CMMS to assist the collection, allocation and compliance of maintenance tasks.
- Applying legal, regulatory and statutory responsibilities to store and maintain equipment and maintenance records.
- Recognising access to consistent up-to-date information helps empower people to do their jobs.
- Maintaining and updating equipment standards confirms that new or upgraded plant or equipment meet current standards.
- Understanding that accurate reporting depends heavily on consistent data structures.
- Recording physical assets in an agreed asset hierarchy allows for effective allocation of costs and supports root cause analysis of equipment failure.

12. Asset Information Management (continued)

Outputs:

- Consistently up-to-date asset management information including asset numbering, technical data, catalogued spares, Bills of Materials, standard maintenance and operating tasks, safety files, drawings, change management, costs etc.
- The development of KPI's for key assets, systems and processes to guide decisions on performance, costs and related risks.
- Information provided in reports is based on a consistent data structure, enabling better identification of improvement opportunities.
- Purchase of plant and equipment is conducted using the latest standards
- Critical and sensitive information is secure and up-to-date
- An auditable trail of asset information

Key Requirements

12.1 – **Defining** and issuing clear **standards for** manage key **asset information**:

- **Establishing a CMMS Data Structure**.
- **Identifying all documentation and formats** required under site management and legislation.
- **Establishing contractor**-owned work history collection and **recording processes** into Glencore Copper systems.
- Preparing **tendering technical specifications** for new equipment or major works.
- Structuring and access for **storing** and updating of server-based **engineering information**.
- Defining **asset performance KPIs** etc.
- **Defining project handover requirements**.
- **Establishing authority levels** and change management processes.
- Establishing processes for **archiving and disposal of information**.
- Using systems **minimise the risk of data loss** and ensure availability.
- **Defining access and ownership of information** generated by contractors, suppliers and OEMs.

12.2 – Establishing correct physical **asset configuration and support information** and making it available to ensure operating, technical and maintenance activities are conducted safely and efficiently:

- Managing information, such as procedures, standards and guidelines.
- **Configuring information** for each asset, such as drawings, manuals, etc.
- **Defining** and implementing **accessibility** requirements.
- **Asset register** and CMMS **configuration** alignment.

12.3 – Establishing and implementing computerised management systems to manage assets:

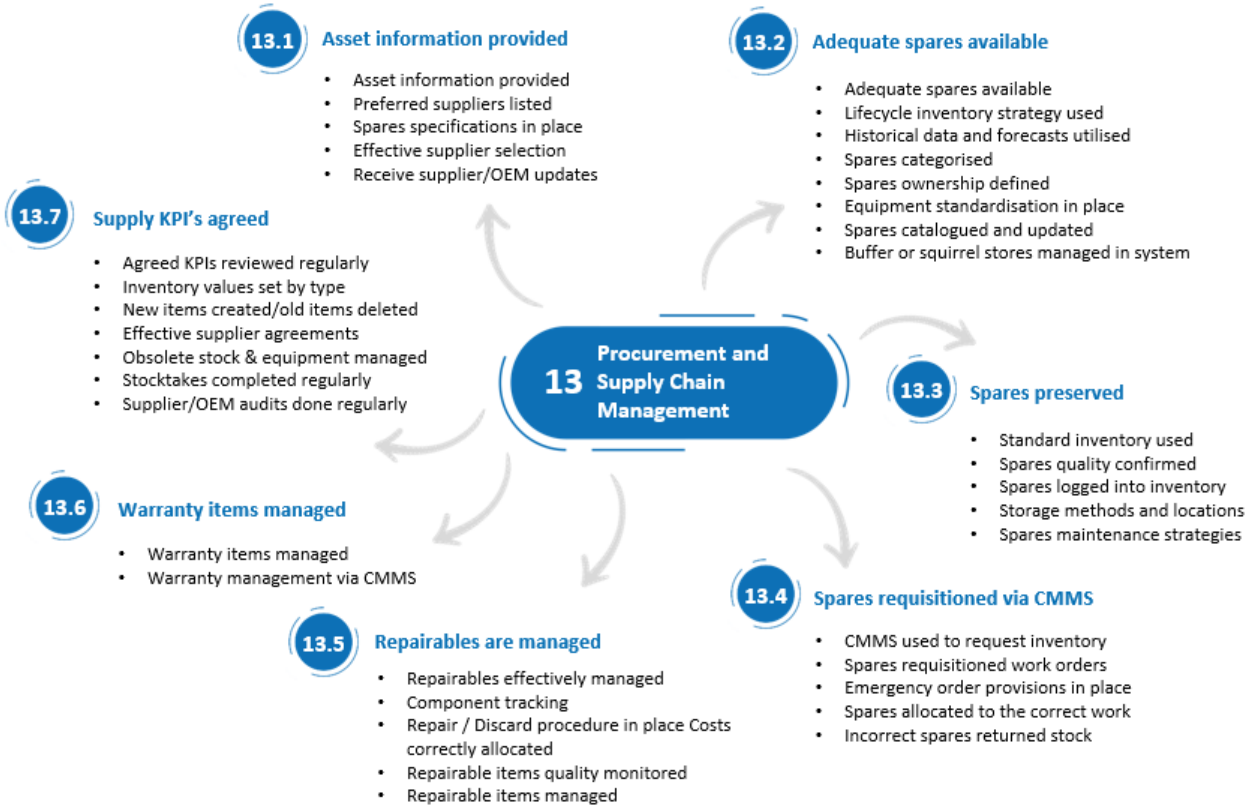
- Implementing the CMMS to a documented standard that meets requirements.
- Establishing systems to communicate equipment and process status, trends and alarms, providing adequate warning and effectiveness reviews.
- Establishing systems to capture delays, losses and downtime reporting.
- Programs are version-controlled and access to modify plant settings restricted.

12.4 – Reviewing regularly the accuracy, completeness and relevance of asset management data so corrective actions are identified and implemented.

13. Procurement and Supply Chain Management

Intent

Procurement and Supply Chain processes and systems are aligned with the Asset Management objectives.



Drivers:

- Realising fully the benefits of the procurement and supply chain management system by integrating with the computerised maintenance management system (CMMS) and each asset's lifecycle plan.
- Linking spares, consumables, energy, water and logistics to the demand requirements of the asset(s) operating and maintenance strategy.

13. Procurement and Supply Chain Management (continued)

- Involving asset owners in reviewing supply holding levels and locations to balance the operational needs and inventory holding costs.
- Identifying that spares and materials availability for critical equipment from OEM/suppliers are important in minimising production downtime.
- Understanding warehousing and storage procedures are key in preventing spares deteriorating during storage.
- Managing asset inventories in line with Glencore Copper's Purchasing and Supply Standards.

Outputs:

- The business is not compromised by quality or availability of consumables and spares.
- Decisions on inventory requirements are made in line with the Equipment Operating and Maintenance Strategy to achieve the business goals at the lowest Life Cycle cost.
- Optimised inventory holdings considers newly acquired assets as well as asset decommissioning and disposal.
- Processes in place for the management of parts and consumables that includes the identification, ordering, issuing and the return of unused items.
- Key indicators that measure the Procurement and Supply Chain performance.

Key Requirements

13.1 – Providing information to procurement on equipment specifications, consumables and spares details that confirm fit-for-purpose spares supplied to the operation:

- **Preferred suppliers**.
- Acceptable **specifications** on brands and standards **for spares**.
- Areas where **spares supply** or holding **agreements** (Consignments, vendor managed, standing orders, forward purchasing agreements) would be beneficial.
- Clear criteria and processes for **effective supplier selection**.
- **Pass on supplier / OEM updates** of consumables and spares details to supply.

13.2 – Setting in place service requirements, inventory levels and location, and suitable contractual arrangements to enable adequate spares availability to achieve asset management objectives:

- Linking the **inventory strategy** to Equipment Operating and Maintenance Strategy **for the whole lifecycle** of the equipment.
- **Historical and forecast** usage **information** is **used** to influence the Reorder Point and the Order Quantity.
- **Categorising spares** (Insurance, critical, consumables, free issue etc.) utilising competent asset management personnel.
- **Defining ownership of spares**, or adequate agreements between sites for shared spares, is in place.
- **Equipment standardisation** guidelines.
- Ensuring **spares** are **catalogued and** prices continually **updated**.

13.3 – Confirming the condition of received, stored and maintained inventory does not influence an asset's operational performance:

- **Spares quality** is **confirmed** before receipt into stock.
- **Spares are logged into inventory** with sufficient information.
- **Storage methods and locations** do not adversely impact the quality of the spares.
- **Spares**, which require maintenance during storage, have **maintenance strategies** implemented in CMMS.

13. Procurement and Supply Chain Management (continued)

13.4 – **Requesting** and withdrawing **inventory** items for maintenance **using the CMMS**:

- **Spares** to be **requisitioned** or reserved through **work orders**.
- **Emergency order provisions** be **applied** when necessary.
- **Spares** withdrawn from inventory to be **allocated to** the **work** and equipment that created the demand.
- **Incorrect** or unused **spares** to be **returned** and re-entered into stock.

13.5 – Managing repairable items through repair orders and competent asset management personnel making scrap versus repair decisions:

- **Repairable items tracked**.
- Economic repair/replace decision process.
- Correct allocation of costs.
- **Repair quality**.

13.6 – **Processing warranty orders** for repair or replacement of failed components under warranty and establishing and eliminating root causes where feasible.

13.7 – **Reviewing agreed** supply performance **KPIs** to confirm time stock delivery, justifying stock levels at site or held by suppliers are acceptable and managing obsolete stock:

- **Inventory value by type** (total, Critical, Slow Moving excluding critical spares).
- **New** items created **and old items deleted** (cataloguing).
- Procurement under contract as a % of total procurement.
- **Effectiveness of supplier agreements**.
- **Sale** or scrapping **of obsolete spares** or equipment.
- **Stocktakes**.
- **Supplier and OEM audits**.

14. Risk Management and Compliance

Intent

The policies and processes for identifying, quantifying and mitigating risk whilst ensuring the asset management activities are compliant with the relevant technical standards and regulatory requirements.



Drivers:

- Risk management is embedded through a structured management framework that establishes a common approach throughout Glencore for identifying, assessing, treating and monitoring business risks.
- The development of HSEC standards, legislative requirements and available standards and guidelines are based on learnings from previous issues, and these recommendations affect most asset management risks.
- Ensuring Glencore's established risk management standards eliminate the need for defining alternative approaches for asset management.
- Ensuring the experience, knowledge and competency required to manage asset risks is clearly defined for each role and incorporated into people management processes.

14. Risk Management and Compliance (continued)

- Frequent updating of legislation, standards and guidelines, as well as the publication of alerts, requires clear processes and roles to enable careful assessment of the applicability to asset management requirements.
- Recognising risk management is an essential process within all asset management frameworks, including ISO 55000 and is one of Glencore Copper's key HSEC standards.

Outputs:

- Incorporating Glencore's HSEC risk assessment standards as part of the asset management framework.
- Asset Management risks are addressed at the appropriate point in the asset management life cycle.
- Significant or recurring risks have been identified and formal practices implemented.
- The techniques for assessing and managing risk for assets have been defined.
- The capability exists to apply efficient and effective asset management risk techniques.
- Conformance with the existing asset risk management approach is monitored and changes in requirements integrated.

Key Requirements

14.1 – Ensuring asset management processes for **risk** assessment and **management** are **developed**, understood and implemented across the whole asset's lifecycle in line with Glencore HSEC standards:

- **Establishing processes** for Failure Modes, Effects and Criticality Analysis (**FMECA**) and Hazard and Operability Analysis (HAZOP).
- Incorporating processes that **assess asset integrity** due to age and or changes to operating practises
- **Applying change management** process.
- Implementing processes for identifying safety critical systems and assessing functional safety.
- Ensure processes for introduction to site.
- Establish operating and maintenance **task analysis and** work instruction **templates**.
- **Creating induction programs** covering key requirements.
- Using guidelines for deferring maintenance or **standing down plant** where maintenance is incomplete.
- **Using processes for handing over plant and commissioning** plant.
- **Establishing permit systems** to perform hazardous activities.

14.2 – **Allocating roles and responsibilities** to review relevant publications to identify any potential asset management requirements:

- **Using Glencore HSEC standards**.
- **Identifying legislated Acts, regulations**, gazetted notices and referenced standards.
- **Using** available **industry and site standards** and guidelines.
- **Reviewing** available **incident reports** (especially High Potential Risk Incidents (HPRI)) and associated alerts and bulletins.

14. Risk Management and Compliance (continued)

14.3 – Developing and implementing documentation (standards, procedures, guidelines and work instructions) to confirm assets are managed in line with mandated and assessed requirements:

- Identifying legislated **management** plans and **standards** of engineering practice.
- Using **site**-specific **standards** of **engineering** practice over **and** above **legislated requirements**.
- **Using equipment**-specific management **guidelines**.
- **Using maintenance** practice specific management **guidelines**.
- Developing equipment strategies to identify maintenance actions considered necessary for compliance.
- Allowing technical specifications **compliance** to be achieved **in procurement**.
- **Risk Assessments for** critical assets for managing the level of risk and controls across the **whole lifecycle**.

14.4 – **Determining the awareness, skills and capability** levels required to assess risk, mitigate that risk and manage compliance across all assets and lifecycles:

- Mapping a minimum level of understanding of requirements against each asset management role.
- **Identifying minimum competencies** for each asset management activity.
- **Adopting** testing and **recertification of competency** for high-risk tasks.

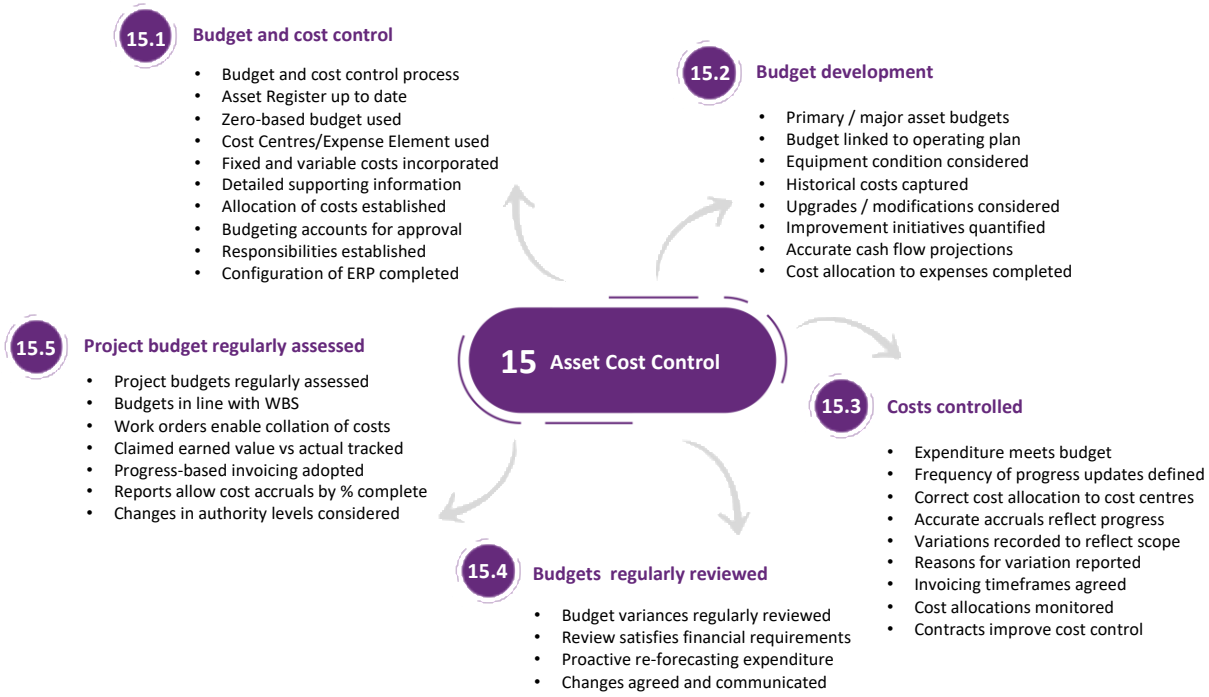
14.5 – Measuring the effectiveness of asset risk management systems regularly and actioning plans to achieve compliance targets:

- **Auditing to measure compliance** with documented requirements.
- Undertaking consultation and feedback on systems efficiency and effectiveness.
- **Reviewing** all **incidents and observations**.
- Using gap analysis and training plans to develop the **capability to manage compliance**.

15. Asset Cost Control

Intent:

Budgets are established and regularly reviewed in sufficient detail to justify the activities required to deliver asset management plans, to allow costs to be controlled and to identify improvements.



Drivers:

- The operation's financial performance is measured and reviewed against budget.
- How assets are managed, have a substantial impact on unit costs, working capital and reliability of sales transactions.
- Recognising asset budgets are an extensive component of the total operations budget, and that accuracy and control significantly affect business's financial performance.
- Establishing zero-based maintenance budgets that (i.e. based on the actual activities and their maintenance types) allow asset managers to make better decisions and capture feedback on real budget change risks.
- Ensuring that maintenance budgets reflect equipment condition, stage of asset lifecycle and operational context or operational goal changes.

15. Asset Cost Control (continued)

- Maintaining cost controls to allow accurate forecasts to be provided to the business.
- Using Budgeting to prioritise funding and highlight target areas for improvement initiatives.
- Developing project budgets allowing cost tracking over longer periods, against progress, allowing work breakdown structures (WBS) and cost accruals.
- Managing asset expenditure in line with financial systems (Delegated Authorities Manual (DAM)/Delegation of Authority (DOA)) in the business.

Outputs:

- Maintenance costs are accurate, understood and under control at all times
- The funding required can be justified with clear benefits and known risks
- Maintenance budgets are zero-based (based on the actual activities and their maintenance types)
- Project budgets are formatted to support the management of project performance
- Clear feedback is given to long term planning based on current performance and future trends

Key Requirements

15.1 – Designing the configuration of the **budget and cost control processes** for effective asset management:

- Formatting maintenance **budget** tools to allow a **zero-based** approach.
- **Asset register**, Cost Centres and expense elements allow adequate categorisation of costs with the level of detail required per asset decided predominately by risk.
- **Incorporating fixed and variable costs** improves the understanding of the drivers of volume and price variations.
- Providing feedback with enough detail to analyse improvements and identify opportunities.
- Establishing cost centres and expense elements for adequate cost categorisation.
- Establishing guidelines for the extent of allocation of costs through work orders and cost centres
- The **budgeting of expenditure** to take in account the timing of the **approval** process within the finance system guidelines and the volume of work and timeframe for execution.
- **Ensuring clear responsibilities** for development, control, reporting and re-forecasting.
- Using **configuration of** enterprise resource planning (**ERP**) finance and maintenance modules to allow adequate asset management reporting either inside or outside the ERP.

15.2 – Developing **budgets** that incorporate forecast operating requirements, asset lifecycle knowledge, historical performance and known improvements – and structured to allow identification of cost driven improvements.

- Establishing a **'line of sight' to the operating plan** and asset management objectives.
- Utilising a forecast operating context (operating patterns and conditions).
- Capturing changes in operation goals.
- Identifying current **equipment condition** and strategy.
- Allocating the equipment by the stage in its lifecycle.
- **Capturing historical costs** and performance.
- Reporting on known **upgrades and modification**.
- Quantifying **improvement initiatives**.
- Using adequate nomination of expense timing for **accurate cash flow projections**.
- Ensuring **correct allocation of cost** to expense type.

15. Asset Cost Control (continued)

15.3 – Monitoring and controlling costs to ensure **expenditure meets budget** or providing clear justification for budget adjustments:

- **Frequency of progress updates** defined for type of work (e.g. Major projects, shutdowns, overhauls, etc).
- **Allocating costs to cost centres correctly**.
- Ensuring **accruals** are **accurate** and reflect **progress**.
- **Recording** purchase orders **variations** and updates to reflect scope changes.
- **Reporting reasons for variation** outside accepted budget tolerance (under/over budget).
- Establishing **agreed timeframes for** approval and rejecting **invoices**.
- **Monitoring cost allocations** effectively via work orders.
- Using **contracts to improve cost control** and management efficiency where applicable.

15.4 – Reviewing and re-forecasting budgets to account for changes to asset performance and operating plans and financial requirements:

- Ensuring **reviews** satisfy the financial requirements.
- Assessing asset performance and **using** it to proactively justify **re-forecast expenditure**.
- Quantifying, **agreeing and communicating the impact of changes** in operating plans.

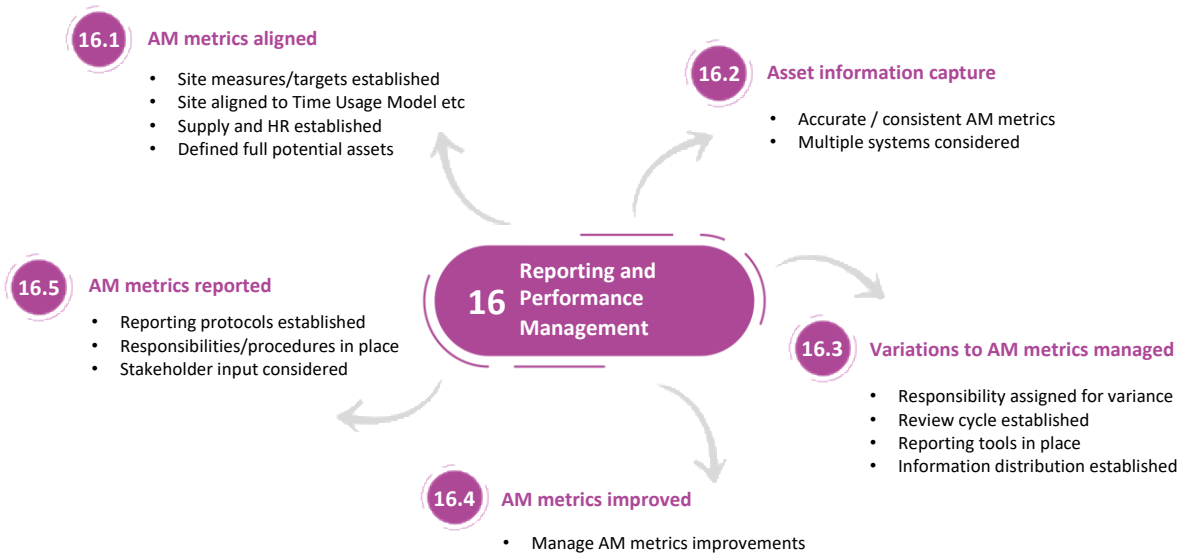
15.5 – **Re-assessing projects regularly** to determine any additional budget and cost control measures required:

- Detailing **budgets in line with** the work breakdown structure (**WBS**).
- Using project **work orders** to facilitate the **collation of costs**.
- Utilising a validation process that assesses **claimed earned value against actual**
- **Adopting progress-based** purchase orders and **invoicing**.
- **Reporting** in formats and at frequencies **to allow cost accrual by percent complete**.
- **Considering changes in authority levels**.

16. Reporting and Performance Management

Intent:

Each level of the organisation to have access to Asset Management information to assess the performance and health of its assets for effective decision-making.



Drivers:

- Giving people the ability to make decisions based on accurate and current information will allow them to perform their jobs effectively and help to create a professional, results-oriented, culture.
- Providing up to date and accurate KPIs and reports to the right level in the business can improve decision-making.
- Information is required to enable problems and opportunities to be identified, prioritised and actions initiated across all levels and functions.
- Information captured should be validated before analysis is carried out and reports are issued.
- Continuous improvement relies on consistency and availability of information.
- Timely and accurate information allows quick feedback on improvements and supports an effective work environment.
- Business requirement to report on Global Metrics.

16. Reporting and Performance Management (continued)

Outputs:

- Asset targets are aligned with the business plan objectives.
- Responsibility for decision-making is clear.
- Decisions add more value because they are made at the right levels in the business.
- Decisions are more cost-effective and timely.
- Performance measures and targets exist and are communicated to all stakeholders.
- Performance measures are consistent across all business units.
- The performance variations due to operating contexts are able to be compared.
- Assets operate in a known condition and with a known capability, and actions to resolve any shortfalls (Planned vs Actual performance) are identified.
- Continuous Improvement projects are well directed because they are based on accurate and relevant information.

Key Requirements:

16.1 – **Establishing** and implementing key **measures and targets** in line with the Glencore Copper Asset Management Metrics Standard:

- **Time Usage Model** (TUM).
- Asset Performance Metrics.
- Work Management Metrics.
- The site to develop additional Asset Management Systems' metrics such as **supply performance**, resource competence, etc.
- **Full potential of the asset** or asset system.

16.2 – Capturing and consolidating asset information through **multiple systems** (with each KPI having one source of the truth):

- The CMMS.
- The process and production systems.
- On board equipment monitoring.
- OEM web portals monitoring equipment health.

16.3 – Assigning responsibility and developing competencies for variation management of Asset Management KPIs:

- Performance management **responsibilities** shall be assigned **for explaining exceptions** and for assigning corrective actions against each Asset Management KPI.
- Clear **review cycles established** with a schedule of meetings, agendas, required attendees, required information inputs and a managed process for actions.
- Required competencies in the use of **reporting tools** and methods of data analysis determined and a program to develop these competencies implemented.

16. Reporting and Performance Management (continued)

16.4 – Developing and implementing a system to **manage improvements to Asset Management** KPIs:

- The actions, target completion dates and responsibilities.
- The improvement opportunity, measurement and tracking.
- Changes and updates to the Asset Management plan(s).
- Sharing of learnings.

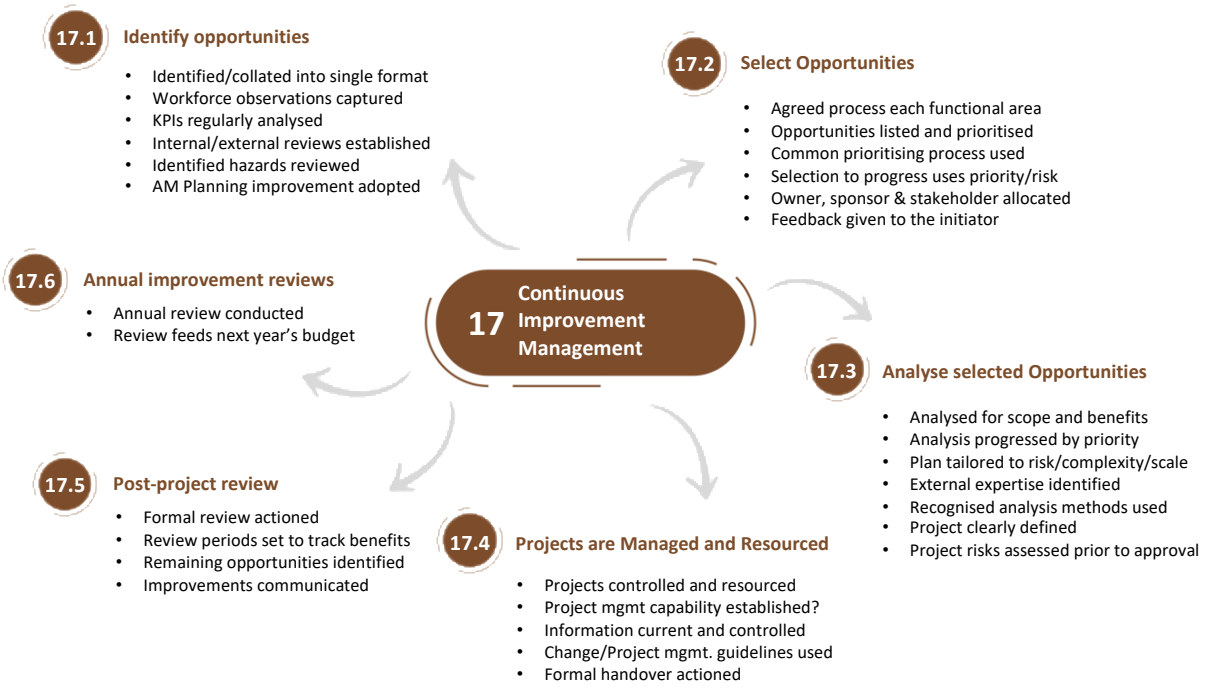
16.5 – Reporting and communicating Asset Management KPIs to employees and contractors:

- **Reporting protocols established** for the timing, format and distribution of reports and other information required by stakeholders.
- **Responsibilities and procedures** to create, distribute and modify information
- Communication forums and meetings are provided to allow **stakeholder input** and actions to be decided such as:
    - Notice boards, newsletters and communication meetings.
    - Start of shift meetings and toolbox talks.
    - Operations planning meetings (Daily, weekly, medium term).
    - Maintenance and Engineering performance meetings.,
    - Budget review meetings.
    - Shutdown meetings.

17. Continuous Improvement Management

Intent:

Processes and systems to build culture that continually identifies, prioritises and implements asset management improvements.



Drivers:

- Prioritising and managing the constant flow of improvement requests asset managers receive from both internal and external sources, ensuring only achievable commitments to beneficial changes.
- Monitoring process performance, regular formal reviews and reporting is undertaken by each organisational level and function to enable the most important opportunities to be identified.
- Involving each organisational level and function in continuous improvement for effectiveness, efficiency and sustainability.
- Understanding most improvement initiatives fail in the implementation phase, so a systematic implementation approach forms part of the system.

17. Continuous Improvement Management (continued)

- Taking a systematic approach to improvement allows long-term planning to estimate and justify the resources level necessary to obtain asset management goals.
- Recognising that continuous improvement as a core activity referenced in ISO 55000 and other asset management frameworks.

Outputs:

- Culture of continuous improvement is embedded within the organisation.
- The cost, resources and competency required to deliver the Asset Management Goals are understood and forecast.
- Benefits and delivery time scale of CI projects are forecast.
- Feedback on the demonstrated value of CI projects is provided and communicated to stakeholders.
- Establish "Best Practice" register for Improvement initiatives are able to be shared throughout the Glencore business.

Key Requirements:

17.1 – **Identifying and collating** all improvement opportunities **into a consolidated format** for consideration:

- **Capturing workforce observations** and suggestions.
- **Analysing key performance indicators regularly**.
- Establishing **internal and external reviews**.
- Using information arising from **identified hazards**.
- **Adopting** business system and process **improvements identified in Asset Management Planning** (Element 3).

17.2 – Utilising for each function a process for registering, prioritising and selecting opportunities to progress:

- **Listing all opportunities** and, as far as possible, all lists are combined.
- Using a **prioritising process** for selection across all opportunities regardless of where it is listed or the source.
- Basing **selection** of opportunities **to progress** on **priority and risk**.
- Setting limits and regularly reviewing which opportunities to register.
- **Allocating** each opportunity to an **owner, sponsor and** identified key **stakeholders**.
- Giving **feedback to the initiator** on the progress/non-progress decision.

17.3 – **Analysing** selected opportunities to determine **the scope** of solutions **and** obtainable **benefits**, and planning in appropriate detail to allow implementation:

- **Progressing analysis and** planning of each opportunity to assigned **priority**.
- **Tailoring** the extent of opportunity planning according to **risk, complexity and scale** as agreed by the sponsor.
- Ensuring that where resource or competency gaps exist, **identified external expertise** assists with analysis to support site personnel.
- **Utilising recognised analysis methods** and standards where agreed to improve determination of the quality and speed of possible solutions.
- **Defining projects clearly** by stating the obtainable benefits, implementation risks, timeframes, resources and required budget.
- **Assessing project risks prior to** implementation **approval**.

17. Continuous Improvement Management (continued)

17.4 – Controlling and resourcing all projects in progress to allow implementation in agreed timeframes and handed over in consultation with stakeholders:

- **Establishing project management capability** and structures.
- **Controlling** project information to confirm secure **current information** is available to the project team.
- Conducting project implementation in line with change management processes and project management guidelines.
- **Actioning a formal handover** when key stakeholders accept the delivered benefits.

17.5 – Reviewing each project after handover to determine success and confirm measures are in place for ongoing performance:

- Setting **review periods**.
- **Identifying the remaining opportunity**.
- **Communicating improvements** to all stakeholders.

17.6 – **Conducting** for each function **an annual review** of delivered improvements and remaining opportunities to re-forecast the required improvement capability **for next year's budget**.

18. Asset Management, Systems Monitoring Audit and Assurance

Intent:

Capable processes and measures are in place to assess the performance, health and effectiveness of the Asset Management System.



Drivers:

- Recognising that improvement in asset performance requires implementation of Asset Management Framework leading practices.
- Measuring and continually improving asset management capability – capability in our people, processes and systems, along with our physical assets, which directly contributes to our aspiration to achieve unrivalled asset performance.
- Increasing maturity in asset management practices increases the likelihood of achieving safety, production and cost objectives.
- Assuring management controls are in place and are being complied with.

Outputs:

- Compliance with ISO 55001.
- Asset management function is responsive and agile so assets continue to meet Glencore Copper's business and strategic objectives and deliver value.
- Adequate and effective Glencore Copper asset management system.
- Stable and predictable production and costs.

18. Asset Management System Monitoring, Audit & Assurance (continued)

Key Requirements:

18.1 – Completing the Asset Management Framework **self-assessed evaluation** (first line assurance) **annually** to provide a baseline for future diagnostic validation.

18.2 – Completing the Asset Management Framework **external evaluation** (second line assurance) **every two years**, facilitated by Group Function, including validation of first line assurance.

18.3 – Self-assessed and external evaluation improvement **opportunities** are incorporated into the site Strategic Asset Management Plan annually.

18.4 – Collating the self-assessment, measurement and validation processes for access by all levels of the leadership team for reviewing the current asset management system performance against the asset management framework intent, allowing for corrective actions.

18.5 – Developing and implementing a performance measured post-assessment business wide action plan considering common strategic themes.

18.6 – **Periodic reviews** of the Asset Management Framework, supporting guiding material and evaluation tools are conducted to ensure alignment with evolving Glencore Copper business objectives.

DOCUMENT CONTROL AND APPROVAL

No amendments to this document may be made without the approval of the General Manager Operational Excellence.

Document Approval:

Position	Name
GM Operational Excellence	Troy Wilson

APPENDIX A – DEFINITIONS

The terms and definitions shown below are applied for the purposes of this Framework and in documents related to this Framework.

Term	Definition
Asset Manager	Any individual or role, which has accountability for the management of physical assets.
Asset Management System	An Asset Management System includes the following: • Asset Management (AM) Framework • AM Standard, AM Guidelines • AM Business Processes and Procedures • AM Diagnostic • AM Strategic Plans • Asset Lifecycle Management Plans • Digital systems • Leaders, Frontline and Organisational Design
Asset Management Framework	Over-arching framework of elements, sub-elements, and practices for Asset Management. Specifies the intent, minimum performance requirements, and guidance for the implementation for each of the Asset Management Framework elements
Competent and Qualified Person	A person engaging in Asset Management activities is deemed competent if they have completed the required competency elements for their role. This may include both Glencore internal competencies and external competencies. A person is deemed qualified if they have completed the required trade, tertiary or statutory competencies as defined by local, regional or national authorities.
Critical Asset	Specific assets that will have a material impact on the business objectives in the event of one or more failure modes. Criticality is assessed in line with the risk management matrix. The impact may be measured in terms of production or financial loss, HSE or License to Operate consequences.
Operating Envelope	A collection of constraints or boundaries on how an asset should be operated, including: • Start-up and shutdown procedures; • Boundary conditions (quality of input/output material such as size, moisture, contamination etc.) • Safe operating limits; • Permissible operating ranges and techniques; • Permissible environmental conditions (e.g., heat/cold, rain, dust); • Specific procedures for abnormal operating conditions.

APPENDIX B – Acronyms

The acronyms and meanings shown below are applied for the purposes of this Framework and in documents related to this Framework.

Acronym	Meaning
5S	Sort, Set in Order, Shine, Standardise and Sustain
AM	Asset Management
AMF	Asset Management Framework
AMP	Asset Management Plan
BOM	Bill of Materials
CAPEX	Capital Expenditure
CMF	Capital Management Framework
CMMS	Computerised Maintenance Management System
ERP	Enterprise Resource Planning
FMECA	Failure Modes, Effects and Criticality Analysis
GM	General Manager
H&S	Health and Safety
HAZOP	Hazard and Operability Analysis
HPRI	High Potential Risk Incident
HR	Human Resources
HSEC	Health, Safety, Environment and Community
ISO	International Organisation of Standardisation
JSA	Job Safety Analysis
KPI	Key Performance Indicator
LOM	Life of Mine
OEM	Operating Equipment Manufacture
OPEX	Operating Expenses
SAMP	Strategic Asset Management Plan
SOP	Standard Operating Procedure
SWI	Standard Work Instruction
TARP	Trigger Action Response Plan
WBS	Work Breakdown Structure
WO	Work Order

Glencore Copper Asset Management Policy



We will manage our physical assets sustainably, without compromising the environment, health and safety of our employees, contractors or the community.

In order to achieve this, we will:

- Embed the Glencore Copper Asset Management Framework at each site, utilising and maintaining the necessary processes and systems to support, control and optimise the application of the Asset Management Framework;
- Create effective Asset Management Strategies that are efficiently executed through the right leadership and organisational structures;
- Ensure our people have clearly defined responsibilities, accountabilities and possess the essential competencies to achieve the Asset Management objectives;
- Plan, operate, maintain and dispose of our assets in conformance with all relevant safety, environmental and regulatory obligations;
- Acquire and/or modify our assets to meet current and future needs of the business and are 'Fit for Purpose' through fully developed operational readiness programs and asset management plans;
- Safely operate and maintain our assets with the parameters that deliver the strategic objectives with maximum efficiency and effectiveness throughout the life of operation; and
- Continuously improve by eliminating losses, minimising waste and mitigating operational risk through root cause analysis and advances in technology.

GLENCORE

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